

Urban Migration, Public Health Amenities, and Local Newspapers in 1870-1940 U.S.

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Abstract

This study investigates newspapers' impact on the migratory decisions of rural households in the 1870-1940 United States, emphasizing the role of information in shaping migration decisions. Contributing to the urban and health economics literature, we offer new insights about how information, specifically newspaper portrayals of public health advancements like water filtration and sewage systems, shapes rural-urban migration patterns. We show that access to information about urban health conditions through newspapers played a crucial role in encouraging rural households to move to cities over the turn of the 19th century. By exploiting a novel linkage between full-count U.S. census data and a unique historical newspaper dataset at the county level, which leverages text-mining and Natural Language Processing (NLP) algorithms, we provide novel evidence that rural households responded to newspaper narratives on public health investments and typhoid occurrences. Rural migrants with access to information about public health investments migrated in higher proportion to sanitation-adopting cities compared to rural migrants with no access to such information, and avoided cities affected by pandemics more than their non-informed counterparts. This finding is consistent with the literature that suggests that such investments significantly reduced mortality from waterborne diseases (typhoid and diphtheria) and improved quality of life, making cities more attractive places to live. As policymakers consider strategies to revitalize urban areas in the post-pandemic era, our study highlights the potential importance of information dissemination in promoting urban growth and development.

Keywords: Urban Migration, Public Health Amenities, Local Newspapers

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1. Introduction

Potential migrants likely do not have all relevant information about each possible destination. This was particularly true for most of history, when search and information acquisition was relatively costly. This friction suggests a potentially important role for the news media in historical migration: learning about the characteristics of other places from the news may change one's beliefs about those places and subsequently the decision of whether and where to migrate.

We study the effects of local newspapers on urbanization via the provision of information about urban amenities. We focus on the U.S. between 1870 and 1940 for two reasons. First, this was a period of rapid urbanization in the U.S.: the percentage of the population living in urban areas increased from 22% to 55%. Second, local newspapers were the only available news media in our setting, as TV had not yet been popularized and radio did not engage in news programming. This setting provides us ample variation in exposure to local news and in urban migration.

The urban amenity of interest to our study is the public health conditions in a city, in particular the incidence of waterborne diseases. Waterborne diseases were an important concern for urban life at the time due to their high death rates, severe symptoms, and their ability to disrupt life and cause massive income loss (Anderson, Charles, and Rees, 2022; Beach, 2022). In 1900, diseases such as typhoid, dysentery, and diarrhea were responsible for nearly 200 deaths per 100,000 residents in the U.S., with most deaths happening during mass outbreaks. Our study period also coincides with a wave of investments into public sanitation projects aimed to eradicate these diseases, such as a water filtration plant or sewage system. Sanitation projects mostly eliminated outbreaks, greatly decreased the incidence of disease, reduced urban mortality. In this paper, we explore the possibility that rural migrants who learned of improved urban conditions due to these projects from local newspapers could have been induced to move there.

We micro-found the location decision of a household using a framework inspired by Berry (1994) and explicitly specify the household's information set to potentially include urban characteristics such as wages, sanitation, and disease rates. Our approach allows for the inclusion of origin-time and destination-time fixed effects and addresses concerns about the endogeneity of the pull-factors of a destination as well as push-factors of an origin. Thus, we identify the role of information in migrants' decision using the variation of migration rates across places with different public health conditions *within* an origin at a given time.

We apply this framework to linked decennial census data that can track migrants between 1870 and 1940 (Abramitzky et al., 2020). Our data on public health conditions include typhoid death rates from Beach (2022) and the implementation of water filtration plants as

well as other public health intervention such as water chlorination, tuberculosis testing, and sewage systems from [Ao \(2015\)](#) and [Anderson, Charles, and Rees \(2022\)](#). To proxy for the availability of information to rural migrants, we use the United States Newspaper Panel from [Gentzkow, Shapiro, and Sinkinson \(2011\)](#) to detect the presence of a newspaper. We directly augment this data with keyword searches on the Chronicling America database from [Dell et al. \(2023\)](#) to measure instances of news coverage of waterborne diseases and changes in public health infrastructure.

Our first set of results demonstrate descriptively that news coverage of a city's public health conditions responds strongly to the implementation of a water filtration project. We present this result across two types of news coverage: direct coverage of the project and coverage of the diseases that result from contaminated water. We find that while newspapers cover water filtration systems around the time of implementation, they mostly do so through the lens of municipal finance and politics rather than public health. We also find that coverage of waterborne diseases (such as typhoid, dysentery, and diarrhea) dropped sharply when a water filtration project was implemented. Furthermore, coverage of diseases vastly outnumbered that of the water filtration projects. This is consistent with previous work by [\(Costa and Kahn, 2017\)](#) suggesting that positive changes in public health conditions filter through the news media as a decrease in "bad news" rather than direct coverage of "good news" for their readers.

Having established that newspaper coverage of a urban destination tracks changes in public health conditions at that destination, we then estimate our empirical model of migration decisions. We find that while migrants across rural origins respond positively to improved urban public health, those coming from origins that have a local newspaper respond at far higher rates. We show that this response is present for across our two proxies of urban health condition being the installation of water filtration or typhoid death rates. These results are robust to the inclusion of origin-time and destination-time fixed effects and so are not influenced by systematic unobserved differences between rural origins with newspapers and those without.

A potential identification concern that rural origins with newspapers might disproportionately respond to public health amenities for reasons other than the provision information: for example, a destination city with a sanitation project can attract migrants from rural origins with a particular skill set, which can correlate with the presence of a local newspaper. We show that our results are robust to pair-level controls that account for demographic congruence between the two places, including that of literacy rates, demographics, and skill sets. We also show that our identifying variation do not predict past migration for the same bilateral pair, which would have been the case if there were substantial pair-level confounders.

We further show that migration decisions not only respond to improved public health

conditions at a destination but also directly to news coverage of that destination. Given concerns about reverse causality in that news coverage might respond with the rate of in-migration from places with newspapers (for example, due to increased demand for news), we use the implementation of a water filtration system as an instrument for news coverage of diseases. As a first-stage, we show that water filtration sharply decreases coverage of waterborne diseases but not other types of diseases. We find that migrants from rural origins indeed differentially respond to changes in coverage of typhoid depending on whether their town has a newspaper.

Related Literature Our paper builds on a literature that found relatively low migration response to local shocks, suggesting the existence of significant barriers to migration (Porcher, 2020 WP). A set of recent papers have found evidence for information frictions to be a major cause (e.g. Wilson (2022) on migration responses to fracking in the U.S., Porcher (2020 WP) on broadband internet in Brazil). In addition, we show that information frictions hinder the extent to which households can migrate in response to changes in amenities (e.g. as in Almagro and Lino (2022 WP) or Diamond (2016)). While we do not provide a model of endogenous amenity investment in our setting, our results suggest that incorporating information can be a valuable addition for future work in that direction.

Our work also relates to a literature on historical urbanization in the U.S. (see Boustan, Bunten, and Hearey (2018) for a survey) and that on the importance of sanitation systems and public health (Beach, 2022; Anderson, Charles, and Rees, 2022; Cutler and Miller, 2005). We show that the implementation of a sanitation system in a city likely induced in-migration from rural areas and that this effect is heterogeneous depending on the information environment experienced by potential migrants.

Finally, we contribute to a literature that studies the role of historical newspaper content in influencing societal outcomes. In particular, we corroborate the findings of Costa and Kahn (2017) that the historical news media focused more on “negative” news of diseases rather than “positive” news of decreased death rate and improved conditions in the city. We extend this finding by showing that the migration decisions of households in fact respond to the absence of disease news, suggesting that households can rationally update on the conditions of the city despite the improvement not being directly covered.

Our paper proceeds as follows. Section 2 describes the data. Section 3 describes the historical background for our study. Section 4 describes our theoretical and empirical model and a discussion of our identification strategy. Section 5 describes our results and robustness tests. Section 6 concludes.

2. Data

A major empirical challenge we overcome in this paper is to measure the salience of health-related events in local newspapers and capture migration responses to differential information flows on destination amenities. We first construct a novel dataset that leverages full-count census data in order to build historical migration flows across U.S. counties and cities. We then pair rural-urban migration flows with data on public health investment and typhoid death rates in cities, as well as a large corpus of digitized local newspaper texts from 1870 to 1940.

2.1. Rural-Urban Migration Flows

This section first describes how we constructed bilateral migration flows from historical U.S. censuses, before drawing key statistics related to rural-urban migrations around the turn of the 20th century.

2.1.1. Building Migration Flows from Historical U.S. Censuses

In order to create a measure of bilateral migration flows in the U.S., we use historical full-count decennial censuses for years 1870 to 1940 from the Integrated Public Use Microdata Series (IPUMS).¹ Each digitized full-count census provides geocoded individual-level data, including details on education, race, income, occupation, family, health and residence. For the period 1850-1940, the full-count censuses contain over 650 million individual-level records.

Census Linking. In order to build intra-U.S. migration flows, we rely on the [Census Linking Project](#) ([Abramitzky et al., 2020](#))² to link individuals between adjacent pair of complete-count census years. Matches between adjacent census years are made based on an individual’s first and last name, age and state/country of birth. In order to limit false positive in migration events, our main results exploit the most conservative matching algorithm from the census Linking Project (ABE-exact), which requires a unique and exact match of name and age within a 5-year age band.³ Our results are robust to the use of the three alternative automated matching methods from the census Linking Projects. Depending on the specific matching algorithm, the linked samples used in our analysis include between 10

¹At the time of this study, the latest full-count census being published for the U.S. is for the year 1950, as full-count censuses have to wait 72 years before becoming publicly available. The 1960s one should be released in 2032.

²<https://censuslinkingproject.org> provides a series of fully anonymous crosswalks between each pair of censuses.

³The approach is described in more detail in [Abramitzky et al. \(2021\)](#).

and 20% of the total U.S. population compared to the full-count census.⁴ Appendix A provides several sampling statistics that support that the distribution of the population in our matched census data is closely aligned to the original full-count census from IPUMS.

Bilateral Migration Flows. We measure migration flows from an origin o to a destination d for an intercensal period (10 years) as the share of individuals living in o at the beginning of the period that we find in location d at the end of the period. Our primary outcome for migration flows is thus equivalent to computing the empirical probability that a person moves from an origin location o to a destination d during an intercensal period t , allowing for people to staying put:

$$m_{odt} = \Pr[\text{household moves from } o \text{ to } d \text{ in period } t]$$

For instance, if o is Mercer county and d is Trenton city in New Jersey, and t captures the intercensal period 1900-1910, m_{odt} captures the share of Mercer county residents in 1900 found in Trenton in 1910.

Rural-Urban Migrants. We define a rural-urban migrant as an individual who moved from a rural county to an IPUMS location identified as a “city”.⁵ Figure 1 plots the distribution of the population in the full-count U.S. census for different type of geographical units (county, city, urban area). The table part of Figure 1 details the share of all individuals from the full-count census living in an area that is not identified as a city. This share decreases from around 80% of the population in 1870 to 50% in 1940.

Figure 2 compares our newly created measure of urban migration flows to the share of population that is urban in the matched full-count census data. The urban population and the urban migration shares are closely related, both in terms of magnitude and time trend, which supports the adequacy of our urban migration measure. Appendix figure A.1 further supports that our matched census dataset is representative of the population distribution of the full-count census, with a correlation coefficient above 0.93 both at the county and city levels. Appendix Section A provides several summary statistics about sampling selection.

⁴Note that the sample only includes male individuals due to the nature of the data, as tracking females through adjacent census decades is complicated by changes of family names post marriage. The census Linking Project thus only provides links for men on the site and our analysis is consistent with the literature using the linked census data.

⁵IPUMS defines a city based on its population: a place is referred to as a city census if it has more than 10,000 contemporary inhabitants in the 1870 and 1880 censuses, and more than 25,000 contemporary inhabitants from 1900 to 1940. Our results are qualitatively similar if we use the less restrictive IPUMS definition of an “urban” county. We prefer the city definition as it allows us to generate a fixed and comparable set of cities across our sample. Appendix F describes in details the definitions of “cities” and “urban/rural areas” used by the full-count U.S. census.

Year	Share of Rural Pop.
1870	0.814
1880	0.784
1900	0.671
1910	0.630
1920	0.576
1930	0.525
1940	0.543

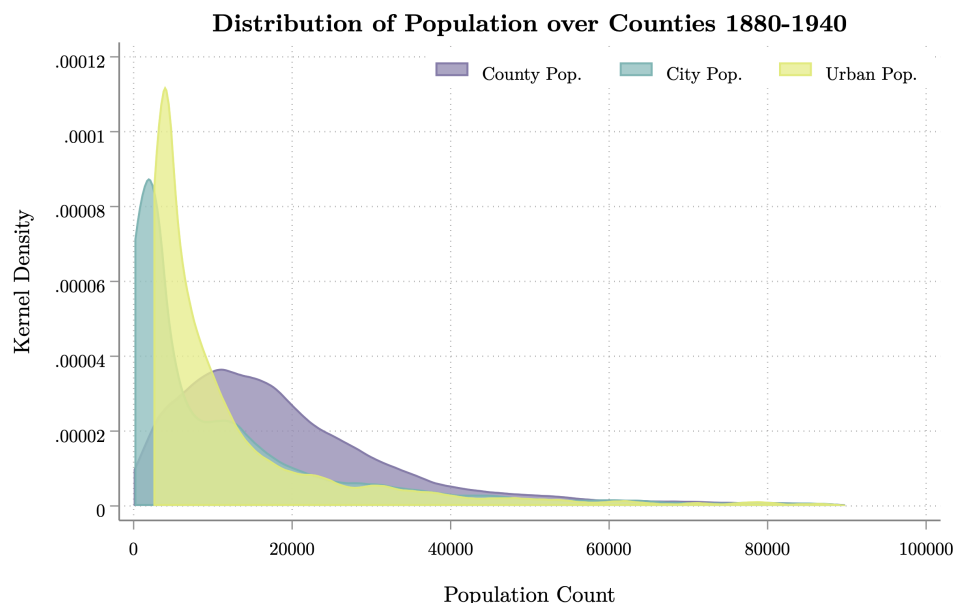


Figure 1: Population by location type in the full-count census data

Notes: Figure 1 is based on the unlinked version of the full-count U.S. census data from IPUMS. A “city” is (mostly) defined as any location with 10,000+ inhabitants. “Urban” areas denote all cities and incorporated places of 2,500+ inhabitants. All areas not classified as urban are designated rural. Households living in smaller, non-urban places are assigned to a state, a county, and a “non-identifiable city” code. The left table details the share of IPUMS Population living in a “non-city” area.

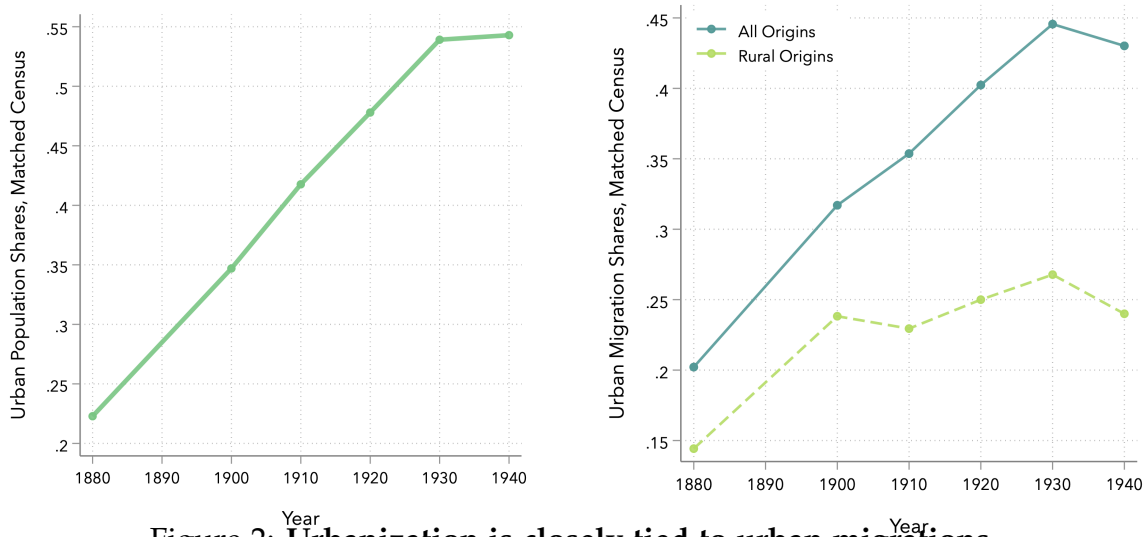


Figure 2: Urbanization is closely tied to urban migrations

Notes: The left figure plots the share of the entire U.S. population living in urban areas, as defined by IPUMS. The right figure plots the share of urban migrants in our data, splitting migrants by origin location type. Both measures are computed based on the linked version of the full-count U.S. census, using the Census Linking Project.

2.1.2. Statistics on Migration Flows

Migration is a local phenomenon. Acquiring information about different potential destinations has historically been costly. It is thus without surprises that migrations in the U.S.

primarily remained a local phenomenon between 1870 and 1940 (Figure 4). The average mover in our linked census dataset only moved 9 miles away from their origin location between two census years, with a standard deviation of 10 miles. This is consistent with the findings of Eckert and Peters (2018) and suggests that information costs may be cheaper in the immediate locality.

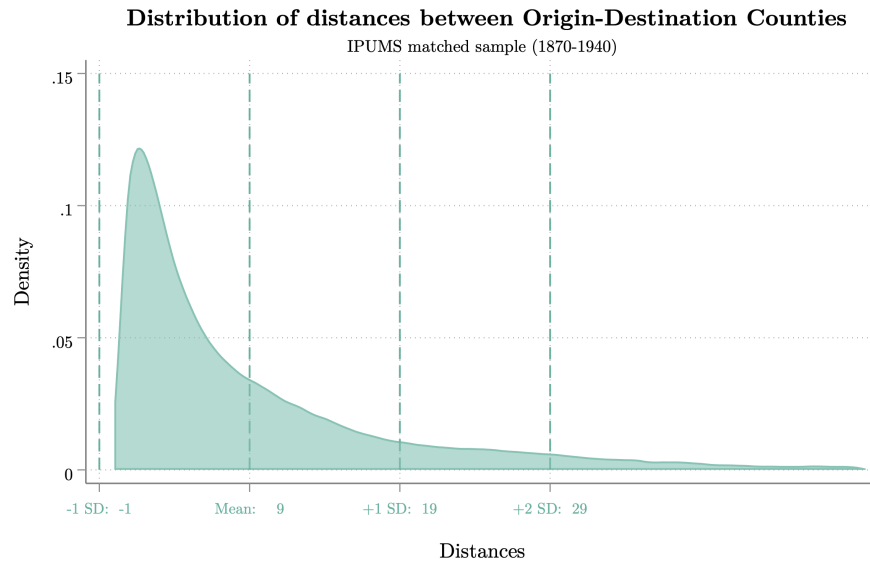


Figure 4: Migrations are mostly a local phenomenon

Notes: Figure 4 reports the distribution of distance between origin and destination counties/cities for individuals identified as movers in our matched census data. The average mover in our linked census dataset only moved 9 miles away from their origin location between two census years, with a standard deviation of 10 miles. Intra-U.S. migrations primarily remained a local phenomenon between 1870 and 1940.

Share of Urban Movers. Migration flows to cities accelerated over the turn of the 20th century (Table 1). Most individuals who moved before 1920 were going from rural areas to urban areas, or from non-cities to cities. Starting around 1920-1930, as more people live in urban counties, this trend reverses and migration flows become mostly urban-to-urban and city-to-city.

Table 1: **Number and Share of Movers to Urban and City Areas, by Year**

Year	Urban Areas	Rural to Urban	Urban to Urban	City Areas	Non-City to City	City to City
<i>Total Number of Movers (Volumes)</i>						
1880	227554	123403	104151	178351	113305	65046
1900	549116	360296	188820	446615	316655	129960
1910	928953	480752	448201	718322	416061	302261
1920	1.420e+06	669103	751136	1.138e+06	639412	499087
1930	1.979e+06	867559	1.111e+06	1.635e+06	872126	762589
1940	1.753e+06	661979	1.091e+06	1.372e+06	636419	735502
<i>Share of Movers (Among all Movers)</i>						
1880	0.253	0.137	0.116	0.198	0.126	0.0723
1900	0.396	0.260	0.136	0.322	0.228	0.0938
1910	0.437	0.226	0.211	0.338	0.196	0.142
1920	0.512	0.241	0.271	0.411	0.231	0.180
1930	0.572	0.251	0.321	0.473	0.252	0.221
1940	0.521	0.197	0.324	0.407	0.189	0.218
<i>Share of Movers (Among the whole IPUMS Population)</i>						
1880	0.0868	0.0470	0.0397	0.0680	0.0432	0.0248
1900	0.176	0.116	0.0606	0.143	0.102	0.0417
1910	0.146	0.0756	0.0705	0.113	0.0654	0.0475
1920	0.169	0.0797	0.0894	0.136	0.0761	0.0594
1930	0.186	0.0817	0.105	0.154	0.0821	0.0718
1940	0.140	0.0527	0.0868	0.109	0.0507	0.0585

Notes: The second (middle) part of the table reports the share of all movers that goes from a given type of origin to a given type of destination (e.g.: total number of movers from rural to urban areas in census year t divided by the total number of movers in year t , for all possible origins-destinations). The share of movers going to urban areas increase over the period, and while the early 20th century migrations were mostly from rural to urban areas, migrations become increasingly urban-to-urban starting around 1920-1930. The third (bottom) part of the reports the share of movers as a percentage of the whole U.S. population from census (for instance, total number of movers to cities in year t divided by the total U.S. population in year t).

2.2. Public health investments

Our dataset on water filtration systems comes from [Ao \(2015\)](#), which records the year of installation of these systems for 74 cities located across 30 U.S. states. The creation of the dataset involves compiling data from historical surveys and censuses of filtration plants published by American Water Works Association as well as supplementary textbooks on water filtration practices. The sample of the cities was conditional on having a population of over 30,000 in 1920, having information on water filtration plant installation dates available, and having existed in the 1880 census.

We supplement the dataset on water filtration with data from [Anderson, Charles, and Rees \(2022\)](#), which provides measures of other public health investments for only 25 US cities. These other investments include water chlorination, clean water projects such as aqueducts, milk standards (such as tuberculin testing for cows and bacteriological standards), and sewage systems.

Table 2 describes these two data sources across counties as well as the characteristics of

those counties. By the end of the 1870-1940 period, 59 cities out of the 74 had installed a filtration system. This represents approximately 1.5% of our county-years.

Table 2: **Descriptive Statistics: Public Health Investments**

	Mean	SD	Description
Public Health Investment			
Sewage Treatment/Diversion	0.024	0.152	= 1 if municipality had a sewage treatment plant or diverted sewage away from drinking water supply, = 0 otherwise
Filtration	0.015	0.123	= 1 if municipality had a water filtration plant, = 0 otherwise
Chlorination	0.033	0.179	= 1 if municipality chemically treated water supply, = 0 otherwise
Clean Water Project	0.022	0.146	= 1 if municipality had completed a clean water project, = 0 otherwise
Bacteriological Standard	0.024	0.154	= 1 if municipality set bacteriological standard for milk supply, = 0 otherwise
TB Test	0.023	0.151	= 1 if municipality required tuberculin testing of cows, = 0 otherwise
County-level Covariates			
Diff. in % Foreign, Dest.	-0.267	0.120	Difference in share of County population foreign born between year t and t-4
Diff. in % Manufacturing, Dest.	-0.210	0.204	Difference in share manufacturing in the County between year t and t-4
Diff. in % Male, Dest.	-0.163	0.128	Difference in share of male population in the County between year t and t-4
Diff. in % White, Dest.	-0.084	0.171	Difference in share of white population in the County between year t and t-4
Diff. in % Literacy, Dest.	-0.515	0.124	Difference in share of literate population in the County between year t and t-4
County $\times$ Year	4122317		

2.3. Newspapers

Our primary data source for the presence of local newspaper dailies is the United States Newspaper Panel from [Gentzkow, Shapiro, and Sinkinson \(2011\)](#). The dataset contains detailed characteristics for newspapers for each election year in the US, including readership size, subscription prices, posted advertising prices, political affiliations, and dates of entry and exit. We use this dataset to construct a measure for the availability of a local daily newspaper for each county in the US during our study period. There were approximately 300 counties in the U.S. with at least one newspaper in 1880 and more than 1,000 of such counties by 1940. Detailed summary statistics of this dataset are available in [Gentzkow, Shapiro, and Sinkinson \(2011\)](#).

Our data on newspaper coverage comes from the Chronicling America database compiled by the Library of Congress ([Dell et al., 2023](#)). To construct a measure of newspaper coverage of urban public health, we use a simple keyword search on the Chronicling America database. For each newspaper page, we detect a “hit” when a phrase related to public health appears near the name of one of our sample cities on the page.

We perform this exercise for two sets of public health phrases. First, we look for mentions of specific diseases. Among waterborne diseases, we include “typhoid,” “dysentery,” and “diarrhea.” We also include a set of “placebo” diseases for which we do not expect to be affected by investments into clean water, which include “plague,” “influenza,” and “tuberculosis.” For mentions of public health infrastructures, we look for mentions of a “water filtration” project.

Our measure of newspaper coverage related to public health is the total count of hits for

each phrase in any given year divided by the total number of mentions for any city that year. This gives us the *share* of mentions of a city that also pertains to the public health topic of interest.⁶

3. Historical Background and Descriptives

In this section we present the historical background that motivates our empirical analysis. First, we describe the state of urban public health in the US and investments in public health infrastructure during our sample period. Second, we describe the relationship between urbanization and public health investments.

3.1. Urban Public Health

For most of American history, cities were deadly places. Important killers of urban dwellers include waterborne diseases which spread via water contaminated with bacteria or parasites such as typhoid, cholera, and dysentery. In 1900, these diseases combined caused approximately 186 deaths per 100,000 inhabitants in the U.S. (Troesken, 2001).⁷

These diseases were salient for reasons beyond their high death rates. For example, consider typhoid which accounted for roughly 2.5% of total mortality during this period-Perfect. (Cutler and Miller, 2005). Infection typically occurs via drinking water tainted with infected feces. Symptoms include muscle ache, diarrhea, severe fever, delirium, internal hemorrhage, and increase the risks of pneumonia and tuberculosis during the disease. The brutality of the symptoms is intensified by their duration: the disease's active period can last from three to four weeks, but the recovery period can be as long as four months. Reinfections were common and there was no effective treatment or vaccine. Furthermore, epidemics were common (Troesken, 1999). Given the severity, length, and frequency of the disease, typhoid was often a signpost for the filthy environmental conditions of the city.

As of the late 19th century, public health officials understood that cleaner water, e.g. via water filtration or chlorination, could eradicate typhoid and other waterborne diseases (Troesken, 1999). This led to a wave of investments in these structures around the turn of the 20th century in the United States (Figure 5). There is an active debate in the academic literature on just how much of the decrease in mortality in the U.S. during this period was due to these investments, with estimates ranging from 20 to 50%. (Cutler and Miller, 2005; Beach, 2022; Anderson, Charles, and Rees, 2022).

⁶For example, if there are 1,000 pages across the database that mention "Milwaukee" in 1900 and there are 5 pages that have "Milwaukee" and "typhoid" in proximity of each other, we compute this share to be 0.005.

⁷For comparison, COVID-19 caused roughly 200 deaths per 100,000 inhabitants in the U.S. between February 2020 to November 2021. (CDC 2021)

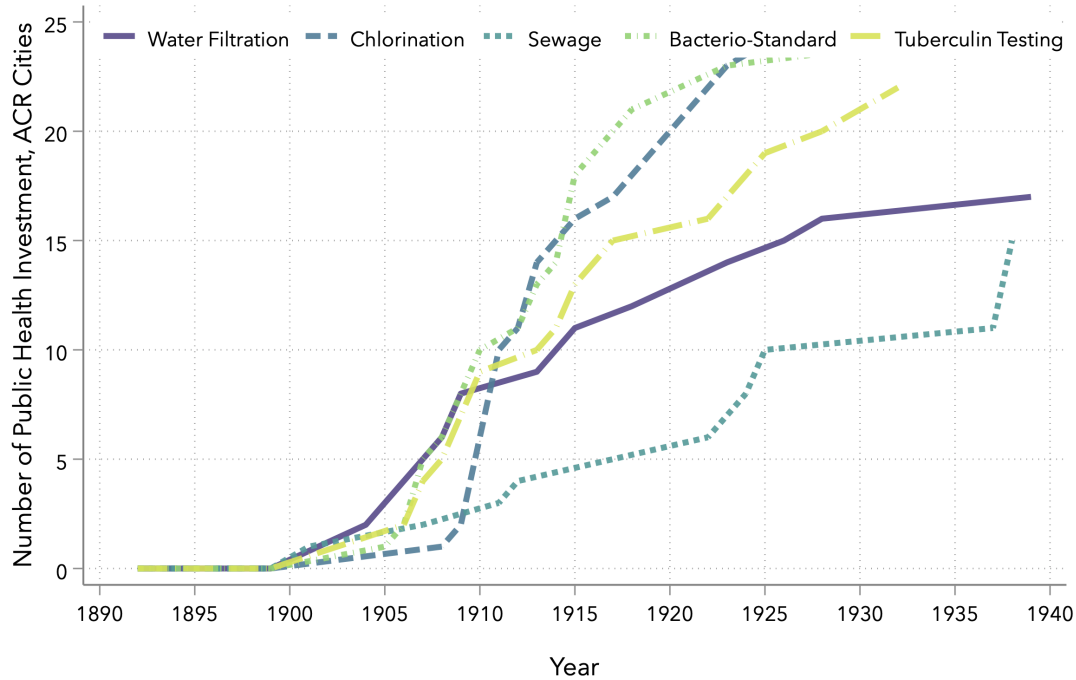


Figure 5: Public Health Investment over the beginning of the 20th century

Notes: This figure reports the cumulative sum of U.S. cities adopting public health investment (PHI) systems in the U.S. The subset is limited to the 25 cities studied by [Anderson, Charles, and Rees \(2022\)](#), splitting the data by type of PHI. The figure emphasizes that most of the changes occurred during the first 20 years of the 20th century. Nearly all cities in the study had adopted bacterio-standard, chlorination and tuberculin testing by 1940. The timing of water filtration and sewage systems was more idiosyncratic, with almost half of major U.S. cities not having implemented such systems by 1940.

3.2. Sanitation Systems and Urbanization

Given the deadliness of the urban conditions, eradicating waterborne diseases was an important goal for cities at the time as they grew. Cities' investment in clean water works closely tracked the rate of urbanization ([Beach, 2022](#)). Municipal ownership of water companies and public investment into filtration systems were viewed as vote-winners by political candidates ([Troesken, 1999](#)). Furthermore, industrialization brought in migrants from the rural area which did not have any immunity to typhoid, increased population density, and was considered a contributor to typhoid epidemics ([Wohl, 1983](#)). Thus, investments into water infrastructure and urbanization were closely linked at the time.

We show that investment in water filtration systems closely tracked urbanization. In Figure 6, we plot the number of cities with water filtration systems from [Ao \(2015\)](#) data against the share of population living in urban counties, which we computed from the U.S. decennial censuses. While the first water filtration systems were only being installed in the 1890s, by 1920, nearly 60 of the cities in our sample had an operational filtration plant. This closely tracks the linear increase in urbanization in the U.S. from 1900 to 1930, which tapered

off between 1930 and 1940.

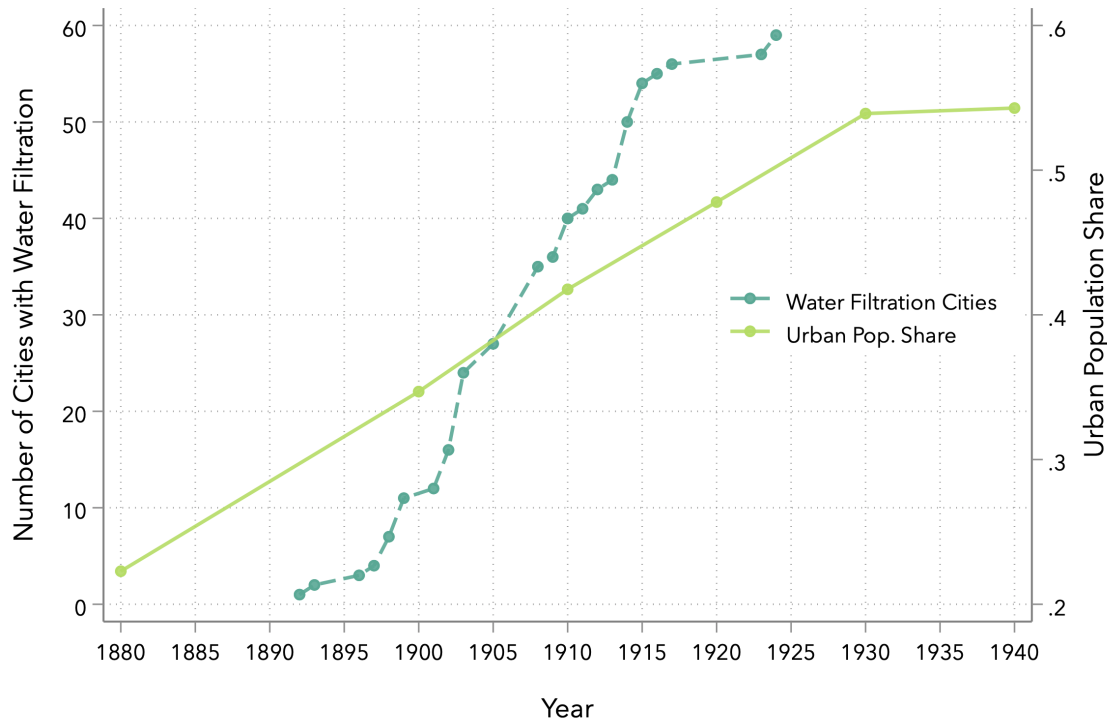


Figure 6: Urbanization & public health coincided

Notes: This figure reports the number of cities with water filtration systems from [Ao \(2015\)](#) data (in dark green) against the share of population living in urban counties (in light green), which we computed from the U.S. census. We find that the timing of urbanization and the implementation of water filtration systems coincided.

3.3. Newspaper Coverage of Urban Public Health

In this section, we present descriptive evidence related to newspaper coverage of urban public health.

The turn of the 20th century witnessed radical changes in the American press. Newspapers became more independent from political parties and more informative ([Petrova, 2011](#)). Increased reach of newspapers also led to an increase in both in-depth investigative coverage as well as sensational journalism ([Gentzkow, Glaeser, and Goldin, 2004](#)). Our informal audit of the newspaper content suggests that news about public health can fit both categories.

First, we find that the vast majority of news coverage related to water filtration projects revolves around their funding. Since these projects were massive municipal projects at the time, there were often political debates and referenda (e.g. on whether to issue municipal bonds) as well as sensational stories regarding government corruption in implementing these projects. Mentions were not always regarding coverage in the newspaper's own city as there

were also discussion of example projects in other cities. Perhaps surprisingly, there was very little coverage celebrating the completion of a project or to advertise the improvement in public health as an amenity.

Second, we find that coverage surrounding diseases at the time broadly fit two types. The first type delivered hard information about local conditions such as case counts, death counts and obituaries. The second type of coverage was more sensational, which included features of urban epidemics and stories of local residents who have a given disease. We note that the majority of disease news was negative and focused on increases in death rates rather than decreases, consistent with [Costa and Kahn \(2017\)](#).

In Figure 7, we report the average share of newspaper articles mentioning either water filtration (top) or waterborne diseases (bottom) in association with a city in our dataset around the time of implementation of a water filtration system. We find that the number of mentions of diseases vastly swamped that of the implementation of a water filtration system. Coverage of water filtration peaks around the time of implementation and shortly after. On the other hand, coverage of waterborne diseases sharply decline after the implementation of filtration systems, consistent with evidence showing that death rates from these diseases also declined after water filtration ([Alsan and Goldin, 2019](#); [Anderson, Charles, and Rees, 2022](#); [Beach, 2022](#)). Disaggregating the second series across waterborne diseases (Appendix Figures C.1) shows that the highest level of coverage (and highest corresponding drop) was due to coverage of typhoid as opposed to dysentery or diarrhea. In Section 5, we show these results formally within a difference-in-differences regression alongside that for placebo diseases.



Figure 7: Capturing News coverage

Notes: These graphs report the share of newspaper articles mentioning either water filtration (top) or waterborne diseases (bottom) that are associated to a city in our dataset. The sample is currently limited to the 74 cities in our public health investment dataset, some of these cities not having an implemented water filtration system over the period. In both cases, we compute the number of keyword hits having both a city name and either filtration or typhoid phrases appearing next to each other in an article. Each graph is centered around the year of implementation of a water filtration system in the city. In the top graph, news related to these sanitation events peak around the time of the public health investment. In the bottom graph, we superpose the share of mentions to waterborne diseases (including typhoid, dysentery, and diarrhea) for these cities. The timing of water filtration investments correspond with a decrease in reports of diseases outbreaks in the newspapers. These negative health shocks are also more likely to peak in the news than positive news about health investment.

4. Model

This section presents our model. First, we describe a simple theoretical framework of location choice that emphasizes the role of information. Second, we use the theoretical framework to motivate our empirical exercises. The goal of the model is to make clear the identification assumptions for our empirical exercises, interpret their results and discuss their limitations.

4.1. Theoretical Framework: Households' Location Choices

This section builds a discrete choice model of household location decisions that emphasizes the role of information acquisition and micro-founds our estimating equations.

4.1.1. Set up

Let $t \in \{1, \dots, T\}$ index periods in time and $\{1, \dots, D\}$ be the set of geographical divisions. For each period t , a household i 's location at the beginning of the period is referred to as its *origin* $o \in \{1, \dots, D\}$ and its location at the end of the period is referred to as its *destination* $d \in \{1, \dots, D\}$. We say that a household i has migrated during the 10-years intercensal period t if $d \neq o$.

Let H_{dt} denote a vector of characteristics of destination location d at the end of the period t that is relevant to the household and observable by the researcher. For expositional purposes, this subsection assumes that H_{dt} is a scalar (e.g. the typhoid mortality rate at a destination). The household may not know H_{dt} and instead have a perception $\tilde{H}_{dt}$ of what H_{dt} may be.

4.1.2. Household's Perception of Location-specific Amenities

We make several assumptions on the household's *perception* of the characteristics of different locations.

First, households have perfect information on the characteristics of the place they currently live in, so that $\tilde{H}_{ot} = H_{ot}$. Second, there are two information regimes possible for each origin location, captured by the indicator C_{ot} for *information coverage*. $C_{ot} = 1$ indicates more information coverage is available at origin (e.g. the presence of a local newspaper). Finally, the perceived location characteristic for a destination $d \neq o$ depends on information coverage in the origin location and can be written as

$$\tilde{H}_{dt} = \begin{cases} \bar{H}_{dt} & \text{if } C_{ot} = 0 \\ \rho H_{dt} + (1 - \rho) \bar{H}_{dt} & \text{if } C_{ot} = 1 \end{cases}$$

where $\bar{H}_{dt}$ is the prior belief about the characteristic of destination d common to households across all origins o . $\bar{H}_{dt}$ thus captures a population-weighted average belief about d over all origin locations o . The parameter ρ captures the extent of updating to the real value H_{dt} for households living in an origin location with more information (i.e. for households living in location such that $C_{ot} = 1$).

Note that this framework nests the strong assumption of perfect updating (e.g. $\rho = 1$ implies that $\tilde{H}_{dt} = H_{dt}$) or completely discarding the information ($\rho = 0$ implies that $\tilde{H}_{dt} = \bar{H}_{dt}$).

4.1.3. Household's Utility from Moving

We specify the location decision of a household i at origin o in period t using the following utility specification for each possible destination d :

$$d_{iot} = \arg \max_{d' \in \{1, \dots, D\}} u_{iodt}$$

$$u_{iodt} = \begin{cases} \tilde{\beta} (\tilde{H}_{dt} - \tilde{H}_{ot}) + X'_{odt} \alpha + \zeta_{dt} - \zeta_{ot} + \zeta_{odt} + \eta_{iodt} & d \neq o \\ 0 & d = o \end{cases}$$

where we normalize the utility of staying put to zero. $\tilde{\beta}$ denotes the moving household's preference over the perceived difference in destination and origin characteristic, $\tilde{H}_{dt} - \tilde{H}_{ot}$. X_{odt} is a vector of migration costs between o and d for intercensal period t . ζ_{dt} is a shifter of utility for destination location d at time t , ζ_{ot} captures the disutility from moving away from location o in period t , and ζ_{odt} is an o - d -pair level shifter of utility. η_{iodt} is an idiosyncratic shock to household i . The household observes ζ_{dt} , ζ_{ot} , ζ_{odt} and η_{iodt} , while the researcher does not.

We can rewrite the utility specification as followed, accounting for the presence of information C_{ot} ,

$$u_{iodt} = \begin{cases} \tilde{\beta} \left[\rho \frac{(H_{dt} - \bar{H}_{dt})}{H_{dt}} \right] C_{ot} H_{dt} + X'_{odt} \alpha + \underbrace{(\tilde{\beta} \bar{H}_{dt} + \zeta_{dt}) - (\tilde{\beta} H_{ot} + \zeta_{ot})}_{\delta_{odt}} + \zeta_{odt} + \eta_{iodt} & d \neq o \\ 0 & d = o \end{cases} \quad (1)$$

where δ_{odt} denotes the mean utility of destination d for households from origin o in period t .

Assuming that the error term η_{iodt} is identically and independently distributed across destinations and households and it follows an extreme value distribution, we can write the predicted share of households from o that migrate to d in period t as a function of the mean utility δ_{odt} ,

$$s_{odt}(\vec{\delta}) = \frac{\exp(\delta_{odt})}{1 + \sum_{d' \neq o} \exp(\delta_{od't})} \quad \text{and} \quad s_{oot}(\vec{\delta}) = \frac{1}{1 + \sum_{d' \neq o} \exp(\delta_{od't})} \quad (2)$$

and obtain the familiar “market share regression” as in [Berry \(1994\)](#):

$$\log(s_{odt}) - \log(s_{oot}) = \delta_{odt}.$$

4.2. Empirical model

Our main estimating equation is directly micro-founded by the discrete choice logit model [1](#) and market share equation [2](#),

$$\begin{aligned} \log(s_{odtg}) - \log(s_{ootg}) = & \beta C_{ot} \times H_{dt} + \underbrace{\alpha_1 \log(\text{dist}_{od}) + \alpha_2 \ln(\tau_{odt})}_{\text{migration costs}} + \alpha_w w_{gdt} \\ & + \underbrace{X'_{odt} \alpha + \gamma_{dt} - \gamma_{otg}}_{\text{location-specific cov.}} + \gamma_{\text{occup}} + \varepsilon_{odgt}. \end{aligned} \quad (3)$$

In this equation, o indexes origins, d indexes destinations, t index time periods, and g index demographic groups (age, race and literacy). The left-hand side captures the log probability of moving from o to d by the end of the intercensal period t , for households belonging to group g . C_{ot} captures information coverage from local newspapers and is a proxy for information available at the origin. In practice, C_{ot} captures as the fraction of period t during which origin location o has a local daily newspaper. H_{dt} is a destination characteristic of interest. In our setting, it will be related to health investments and either measure (i) the fraction of time during period t that destination d has implemented a water filtration system, or (ii) mortality due to typhoid outbreaks. The interaction terms represents preference over public health amenities, while the coefficients α_1 and α_2 captures a moving cost

X_{odt} is a vector of origin-destination-time specific covariates and γ_{occup} is a work occupation fixed effect. ε_{odtg} is a regression error term. The γ terms represent preferences over various location characteristics: γ_{dt} and γ_{otg} are destination-time and origin-time-occupation group fixed effects respectively. The origin-year fixed effect ensures that we are comparing two destinations d and d' for a given origin o , while adding a destination-year fixed effect

ensures that what is left in the unobserved term is changes in moving costs from o to d over time. Location-time specific control variables include share white, male, foreign-born, literate and share working in the manufacturing sector. Each control variable is measured as the difference between origin and destination locations.

In our main reduced-form specification, we will allow for a vector of perceived amenities that vary with information at origin. First, we include a measure of health at the destination, such as death rate from typhoid and the presence of a sanitation infrastructure. Second, we include a measure of wages at the destination. Sections 2 and Appendix Section F detail our construction of these variables.

The parameter of interest is β , which is a product of three distinct objects,

$$\beta = \tilde{\beta}\rho \frac{(H_{dt} - \bar{H}_{dt})}{H_{dt}}$$

In our model, $\tilde{\beta}$ represents the direct value of a characteristic on a household's utility, ρ is the household's extent of updating of their beliefs given the information available in C_{ot} , and $\frac{(H_{dt} - \bar{H}_{dt})}{H_{dt}}$ is the extent of correction of the household's belief given their prior if they fully discard their prior. We cannot separately identify any of these objects without data on the other two. We will instead present a range of possible values for each of these objects given our estimate of β and calibrated values for the remaining ones.

Our use of fixed effects γ_{dt} and γ_{otg} directly corresponds to objects in a household's utility that is constant at the destination-time level ($\tilde{\beta}\bar{H}_{dt} + \xi_{dt}$) and those at the origin-time level ($\tilde{\beta}H_{ot} + \xi_{ot}$). As in [Berry \(1994\)](#), our regression error term corresponds to the origin-destination-time-occupation level unobservable characteristic ξ_{odtg} .

4.3. Identification assumptions

Our primary identification assumption is that the co-variation in origin information environment (e.g. the presence of a newspaper) and the destination characteristic of interest (e.g. typhoid rate) is exogenous to the flow of migrants between those two places after controlling for unobservable characteristics specific to each place. In particular, any variation in the flow of migrants from o to d that correlates with origin information and destination characteristic other than through preference for that characteristic and belief updating violates this assumption. Formally, we assume that

$$\mathbb{E} [\xi_{odtg} | C_{ot}H_{dt}, X_{odt}, \gamma_{dt}, \gamma_{otg}, \gamma_g] = 0.$$

Importantly, we do not assume that the origin information environment is exogenous to

migration or that the destination characteristic is exogenous to migration. In order to make this discussion concrete, let us consider C_{ot} as the presence of a local newspaper at the origin and R_{dt} as the presence of a sanitation structure at the destination.

We do not assume that a local newspaper’s presence does not correlate with migration. In fact, entry of local newspapers is highly correlated with the predicted profit at that market and thus population growth. We only require that it does not differentially influence migration to places with sanitation structures as opposed to places without other than through preference for sanitation and updating beliefs about the presence of sanitation structures. To the extent that the presence of local newspapers correlates with unobservable characteristics specific to that origin location in that time period, we can control for those unobservable with the use of origin-time fixed effects.

Similarly, we do not assume that a sanitation structure’s presence at the destination to be exogenous to in-migration. Sanitation structures are more likely to be implemented at cities that are able to fund large public works projects and in the aftermath of deadly outbreaks and we are able to control for these factors with the use of destination-time fixed effects. We only require that it does not differentially influence in-migration from places with a local newspaper as opposed to places without other than through preference for sanitation and updating beliefs about the presence of sanitation structures.

5. Empirical estimates

5.1. Fixed-Effects Regression

Table 3 reports the results from reduced-form equation 3. All regressions include interactions of distances and water filtration or typhoid death rates, interactions of distances and news coverage, log transportation costs (from Donaldson and Hornbeck (2016)) and median wage at the county level. County-level controls include share of population that is male, white, literate manufacturing and foreign.

For our preferred specification with controls for demographic congruence (Column 4), Given an origin o without a newspaper, on average the share of residents from o moving to d with water filtration would increase by 36% compared to that for a destination d' without a filtration system. Origins with the presence of a newspaper experience a 70% additional increase in migration toward destinations with a filtration system relative to origins without a newspaper. Taken together, origin-destination pairs where the origin has a newspaper and the destination has a water filtration system experiences on average 105% greater migration share than other pairs. Including destination-year fixed effects (Column 8) shows that origin-destination pairs where the origin has a newspaper and the destination has a water filtration

system is only 36% higher than other origin-destination pairs on average. We interpret this as strong evidence that while rural migrants overall respond to improved public health conditions in an urban destination, there are strong differential responses by migrants with access to information about such conditions.

Similarly, in columns 1 and 2, we establish that origin-destination pairs where the destination has higher typhoid rates experience lower share of migrants than other pairs. In addition, there are once again strong differential responses by migrants from origins with newspapers, suggesting that changes in public health conditions are more salient for these rural migrants.

Table 3: Two-ways Fixed Effect Regressions of Migration Flows on Typhoid or Water Filtration News Coverage

FEs:	<i>Dependent Var.: Probability of moving</i>							
	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage × Typhoid rate (deaths by thousand)	-1.283*** [.4321]	-.06172 [.3512]			-.7668*** [.2735]	-.4549* [.2603]		
Typhoid rate (deaths by thousand)	-.4293* [.2335]	-.09899 [.307]						
News Coverage × Water Filtration			.65** [.2904]	.6944*** [.2161]			.4444*** [.1578]	.3581** [.1491]
Water Filtration			.6702*** [.1931]	.3636*** [.1194]				
Log Distance	-.8238*** [.1043]	-.8686*** [.09894]	-.6164*** [.07919]	-.7361*** [.06694]	-1.028*** [.02637]	-1.016*** [.03157]	-.9253*** [.01435]	-.9306*** [.01656]
Mean wage of group g in d	2.0e-07*** [3.9e-08]	8.7e-08*** [2.6e-08]	2.7e-07*** [4.8e-08]	2.1e-07*** [4.0e-08]	5.4e-08*** [1.3e-08]	4.6e-08*** [1.5e-08]	6.2e-08*** [7.7e-09]	5.2e-08*** [8.8e-09]
Log Distance × Water Filtration	-.06374 [.08663]	-.06173 [.06835]	-.2952*** [.03719]	-.2701*** [.0335]	-.1113*** [.02764]	-.1125*** [.03138]	-.1668*** [.02148]	-.1624*** [.02414]
Log Transportation Cost - Donaldson Hornbeck	.6119** [.256]	.4868* [.2762]	.5505*** [.1952]	.6332*** [.1871]	.7022*** [.05867]	.6561*** [.06896]	.778*** [.03786]	.7693*** [.04385]
Share White (D-O)		-4.316*** [.9655]		-3.696*** [.5019]		.03662 [.2662]		-.08264 [.4434]
Share Male (D-O)		.3063 [3.762]		-3.024 [1.869]		3.404 [2.338]		5.68** [2.252]
Share Manuf. (D-O)		-5.366*** [1.444]		-1.549* [.8049]		-.5133 [.4642]		-.7896 [.5183]
Share Foreign Born (D-O)		7.273*** [.7045]		5.936*** [.5874]		.2411 [1.035]		.2048 [.7616]
Share Literacy (D-O)		15.76*** [3.026]		12.78*** [1.639]		2.782 [3.191]		3.943 [2.482]
Occupation FE	✓	✓	✓	✓	✓	✓	✓	✓
Constant	-4.227*** [.2678]	-4.59*** [.2657]	-5.548*** [.2022]	-5.736*** [.1896]	-4.155*** [.05777]	-4.137*** [.0843]	-5.346*** [.04683]	-5.348*** [.05768]
R2	.5591	.6572	.6008	.6647	.8066	.8087	.8135	.8149
F-Stat	315	336.1	616.9	491.9	813.4	490.4	1085	667.8
Dep. Var.: Mean	-4.76	-4.77	-5.381	-5.42	-4.76	-4.77	-5.381	-5.42
Dep. Var.: Std. Dev.	1.522	1.536	1.642	1.658	1.522	1.536	1.642	1.658
N Obs.	4.1e+05	3.2e+05	9.6e+05	7.7e+05	4.1e+05	3.2e+05	9.6e+05	7.7e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: All coefficients are standardized beta coefficients. Standard errors are clustered at the origin-group-year and destination-year level. Includes interaction of distance and PHI, distance and news coverage, log transportation costs ([Donaldson and Hornbeck, 2016](#)) and median wage. County-level controls: share male, white, literate, manufacturing and foreign.

5.2. Migrants Respond to News Coverage

In this section, we augment the results of the previous section to highlight that when a destination city implements a water filtration system and experiences a decline in typhoid rates, rural migrants from origins with newspapers differentially respond to the associated decline in news coverage of typhoid by moving to that destination.

Using our measure of news coverage constructed in Section 2.3 as a regressor within our primary estimating equation 3 presents two primary challenges.

The first challenge is a data limitation: we only observe disease coverage for large urban destinations but not for all destination places (such as potential rural destinations or places not named in the census). We take two approaches to this problem. First, we impute the missing amount of coverage to be 0 for rural areas, while unnamed urban areas gets the average coverage measure for cities without typhoid outbreaks/PHI investments during that period and within the same state. We believe that this is justified: the rate of typhoid in rural areas was very low (and likely did not receive any coverage). For robustness, we present qualitatively similar results in the appendix where we limit our sample to only urban areas where we have data on news coverage of diseases.

The second challenge is related to pair-level reverse causality. News about diseases at a given destination d should correlate with mentions of this destination in a newspaper at the origin o , since migrations from o to d would increase demand for news of d at o . Due to this problem, we instrument for coverage of typhoid by using the implementation of water filtration systems in each city. Our assumption is that the implementation of a public health investment was exogenous to the differential flows of in-migration from origin towns with newspapers and those without.

Figure 8 presents the first stage results for this instrumental variable strategy. We plot the normalized number of hits for "typhoid" linked to a city name per thousand articles around the opening of water filtration plants at time 0, with the x-axis denoting the number of years from the date of implementation. Investments in filtration plants coincide with a sharp drop in the share of news mentioning typhoid related to the city. This is consistent with the disease being gradually eradicated and leading to less news coverage overall (with F-stat of more than 1,000). The effect is gradual and pronounced, consistent with works showing a similar trend effect in the eradication of the disease itself. (Beach, 2022; Alsan and Goldin, 2019).

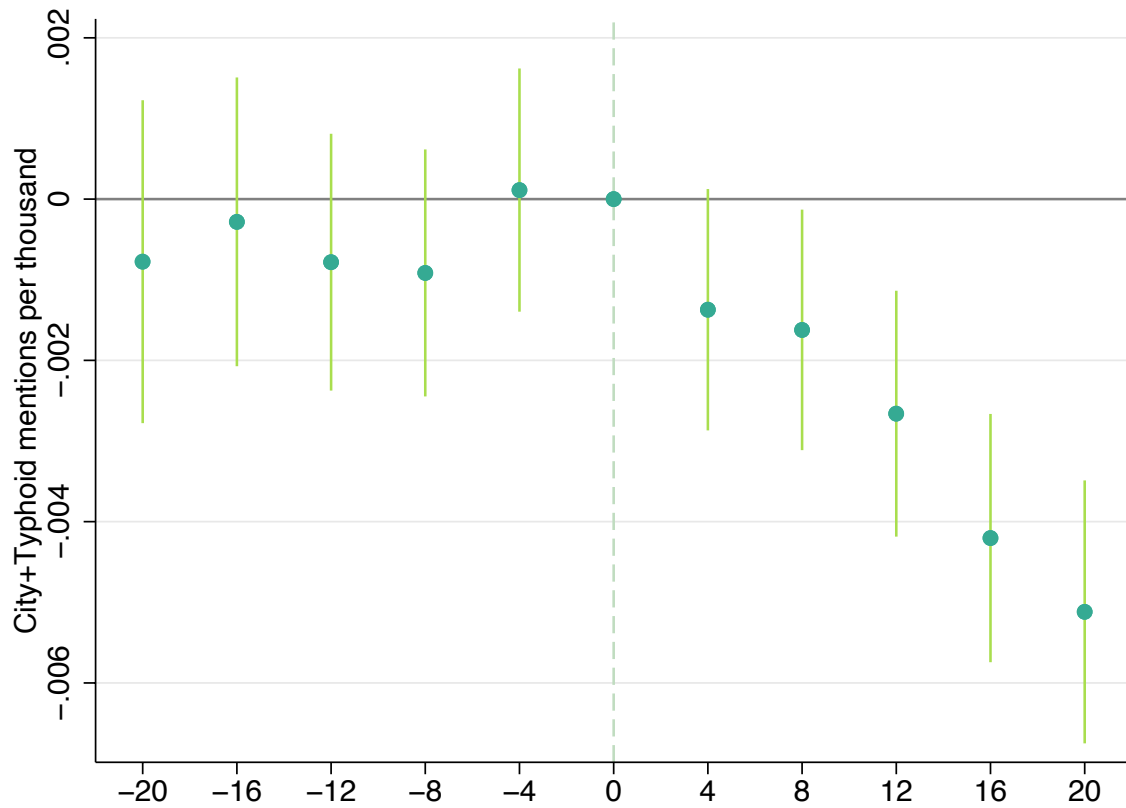


Figure 8: Effect of water filtration system on typhoid news coverage

Notes: This figure presents the changes in newspaper coverage of typhoid in response to the implementation of a water filtration system. The y-axis denotes the number of hits for a destination city and “typhoid” on a newspaper page divided by the total number of hits for a destination city during that time period. The x-axis denotes years since the implementation of the city’s water filtration system, grouped into four-year periods.

Table 4 describes our second-stage results. When we do not instrument for typhoid news (Columns 1 and 2), we find that origins with a newspaper have more migrants toward destinations that have a higher incidence of typhoid news. This likely represents the previously discussed bias: newspapers are more likely to report on places that experience a high rate of in-migration, especially from places where there are local newspapers. When we instrument for the amount of coverage of typhoid with the implementation of a water filtration system, we find that migrants from origins with newspapers respond especially negatively to the increase in coverage of typhoid news. In our preferred specification (Column 4), an increase in one standard deviation of typhoid news coverage of a destination leads to a 61% lower migration share from an origin with a newspaper.

Table 4: News Coverage Instrument

	OLS		IV	
	(1)	(2)	(3)	(4)
Newspaper Presence $\times$ Typhoid News (std.)	0.085*** [0.014]	0.11*** [0.021]	-0.50** [0.20]	-0.61** [0.27]
Newspaper Presence $\times$ Urban/rural dest.	0.050*** [0.018]	0.039* [0.022]	0.84*** [0.25]	0.83*** [0.29]
Log Distance	-0.98*** [0.0073]	-0.99*** [0.0087]	-0.98*** [0.0078]	-0.99*** [0.0095]
County Covariates, in Diff. (D-O)		✓		✓
Origin-Year FE	✓	✓	✓	✓
Destination-Year FE	✓	✓	✓	✓
R2	.756	.758	.5386	.5416
Dep. Var.: Mean	-7.00	-7.02	-7.00	-7.02
N Obs.	2.7e+06	2.0e+06	2.7e+06	2.0e+06

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table describes results for our regression of migration flow on the typhoid news coverage. Columns 1 and 2 describe the ordinary least squares specification without any instrument. Columns 3 and 4 describe the two-stage least squares specification where the instrument is the implementation of a water filtration system at the urban destination. Each column includes origin-year and destination-year fixed effects, and columns 2 and 4 include controls for demographic congruence between the origin and the destination. We report coefficients for three variables. The first is the interaction term between whether the rural origin has a newspaper with the standardized amount of typhoid coverage of a destination. The second is the interaction term between whether the rural origin has a newspaper and whether the destination is classified as a rural or urban place in the census. The third is the natural logarithm of the distance between the origin and the destination. All standard errors are clustered at the origin-year and destination-year level.

In order to ascertain that changes in migrants' decisions are due to changes in waterborne diseases and not other, unrelated general health conditions of the city, we replicated the first-stage of this instrumental analysis for a set of diseases likely unaffected by the implementation of a water filtration plant: plague, influenza and tuberculosis. Figure 9 reports the resulting event study graphs. The pronounced decrease in coverage of waterborne diseases reported in 8 is replaced by an increase in coverage of other events. We note that since migrants respond to the implementation of a water filtration plant by increasing their immigration, it's plausible that readers perceive pulmonary diseases as less severe, less deadly, or less sensational.

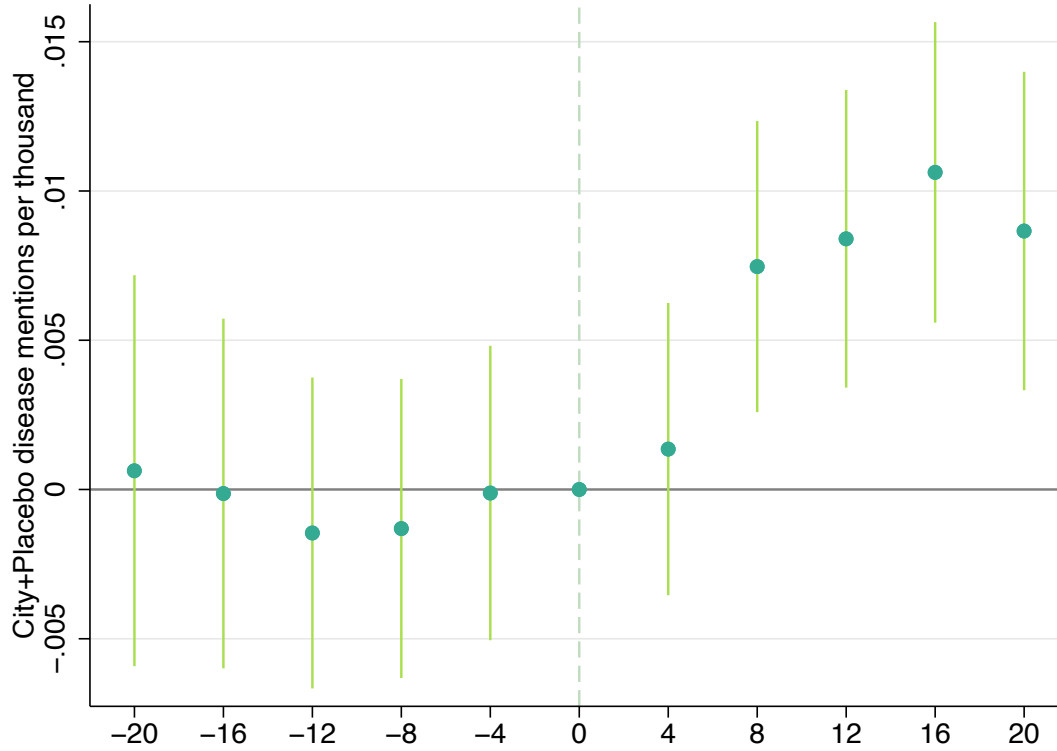


Figure 9: First stage of PHI on destination placebo disease news

Notes: This figure presents the changes in newspaper coverage of tuberculosis, influenza, and plague in response to the implementation of a water filtration system. The y-axis denotes the number of hits for a destination city and tuberculosis, influenza, and plague on a newspaper page divided by the total number of hits for a destination city during that time period. The x-axis denotes years since the implementation of the city's water filtration system, grouped into four-year periods.

6. Conclusions

This paper demonstrates the significant role that daily newspapers in rural areas played in the process of urbanization in the United States between 1870 and 1940. Through the analysis of a unique dataset that combines full-count U.S. census information with historical newspaper archives, we provide compelling evidence that the dissemination of information about urban amenities, particularly public health improvements such as water filtration and sewage systems, directly influenced the migratory decisions of rural households. Our findings highlight the critical impact of local newspapers in bridging the informational gap about the benefits of living in cities that were investing in sanitation infrastructure, thereby making these urban centers more attractive to potential migrants. Notably, we observe that rural migrants were significantly more likely to relocate to areas boasting public health investments (PHI) when they had access to a local newspaper that reported on these urban advancements.

The implications of our study extend beyond the historical context of the United States,

offering potential insights into the ongoing urbanization trends in regions such as Latin America, Asia, and Africa. These areas are currently experiencing rapid urban growth in mega-cities like São Paulo, Jakarta, and Mumbai. One plausible explanation for this phenomenon, as suggested by our research, is the increased availability of information. In today's digital age, where information dissemination is even more widespread and accessible than in the past, our findings underscore the importance of information availability in shaping migration patterns. This is particularly relevant in the context of pandemics, where real-time information about public health conditions can significantly influence the reallocation of populations.

As the world continues to urbanize at an unprecedented rate, the lessons drawn from our study emphasize the need for policymakers to focus on enhancing information dissemination about urban amenities and public health investments. This strategy could be instrumental in promoting urban growth and development, especially in developing regions where the rapid expansion of cities presents both challenges and opportunities. In conclusion, our research not only contributes to the academic discourse on spatial and urban economics but also offers practical insights for contemporary urban planning and policy-making, highlighting the enduring relevance of information in shaping human migratory behaviors.

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Appendices

A. Sample Selection: Sampling from Censuses

A. Over/Under Sampling of Population in Matched Census

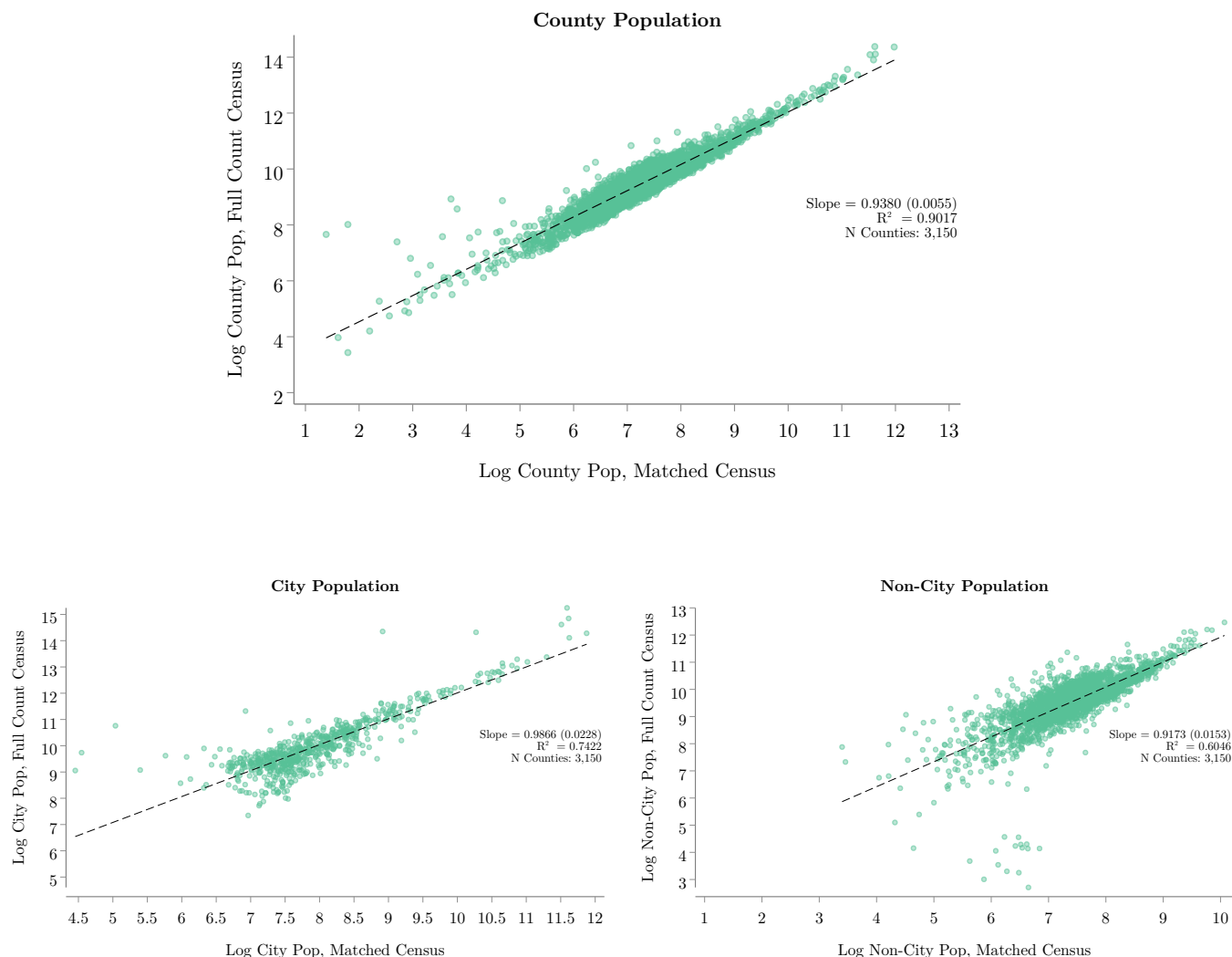


Figure A.1: Population Distribution in Matched versus Full-Count U.S. censuses

Notes: These graphs plot county (top), city (bottom left) or non-city (bottom right) level population (in logs), comparing the distribution of population in the full-count U.S. census (1870 to 1940) to our matched version of this full-count census. The matched version is a subset of the full-count census in which individuals have been matched between two census years. We use the most conservative matching method from [Abramitzky et al. \(2020\)](#) to match individuals in adjacent census, using unique exact name and age in a 5-year age band.

County Population: Over/Under Sampling in Matched Census

Average over 1870-1940

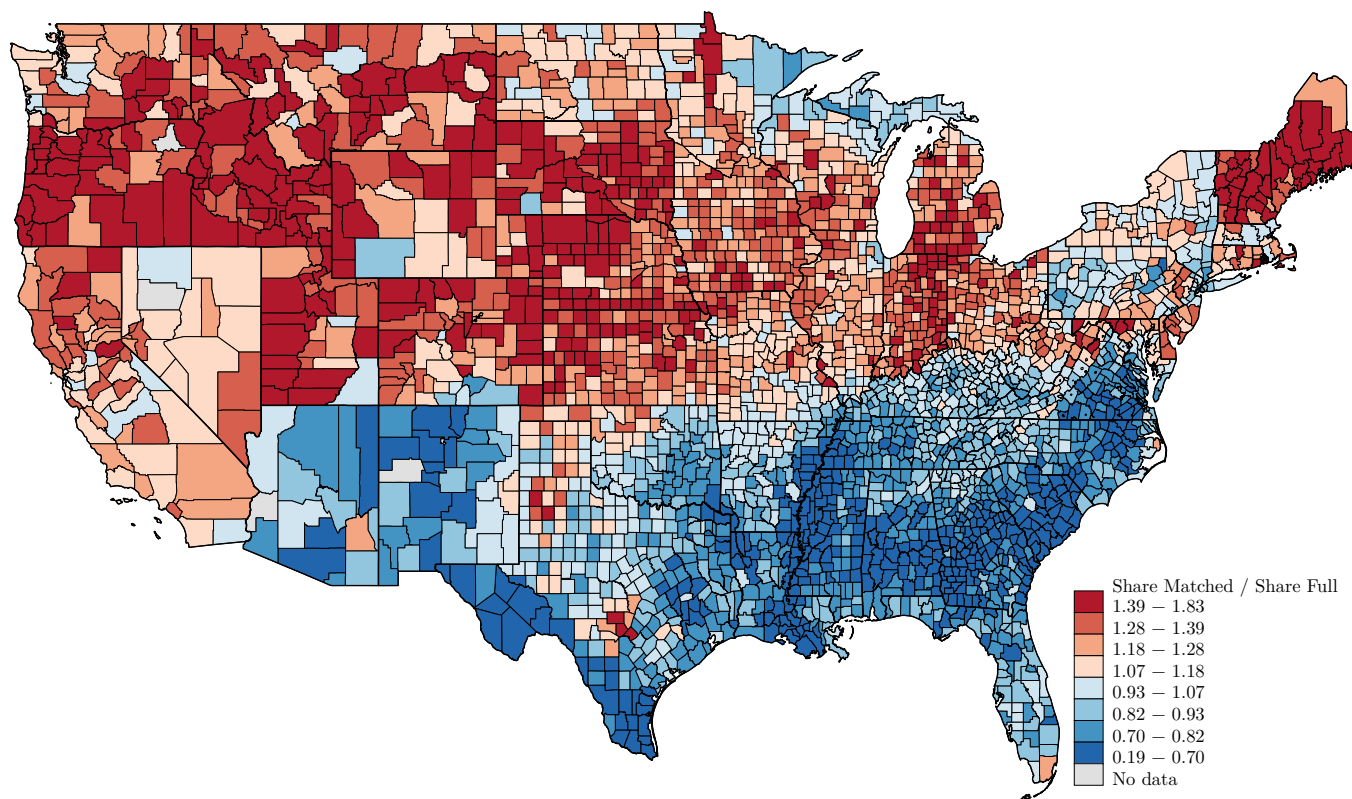


Figure A.3

Notes: This map compares the distribution of county-level population in the full-count U.S. census to our matched version of this full-count census. County populations are averaged over the period 1870 to 1940. The matched-census version is a subset of the full-count census in which individuals have been matched across two adjacent census years, in order to compute migration responses. We use the most conservative matching method from [Abramitzky et al. \(2020\)](#) to match individuals in adjacent census, using unique exact name and age in a 5-year age band (ABE-exact). For each county, we compute the population share compared to the entire U.S.. We then divide the population share of each county in our matched census data by the population share in the full-count census. Counties in red are oversampled in our matched census dataset compared to the full-count census, while counties in blue are undersampled. The South of the United States is more likely to suffer from undersampling, which may be largely driven by higher shares of African-American population in the south. The Census Linking Project mostly captures white males and African-American populations are less likely to be matched across census years (see [Abramitzky et al. \(2020\)](#) for more details).

B. Population Distribution, by County

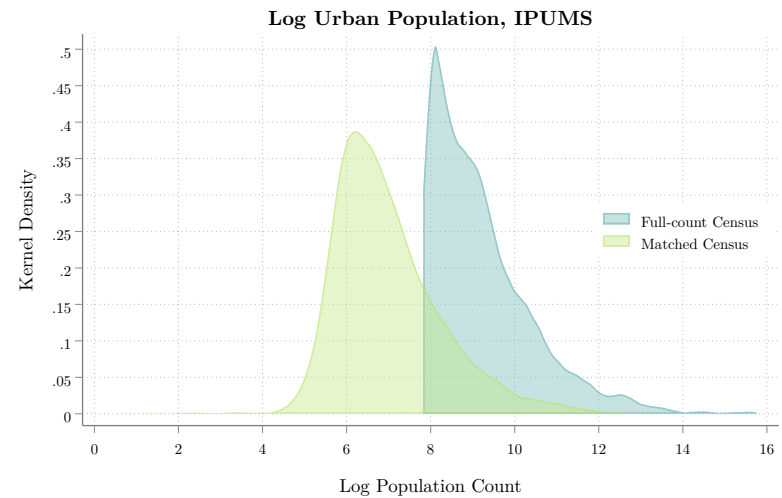
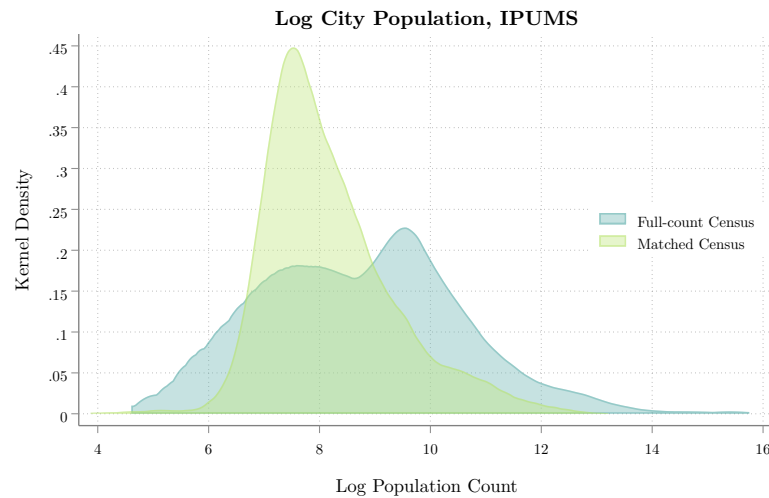
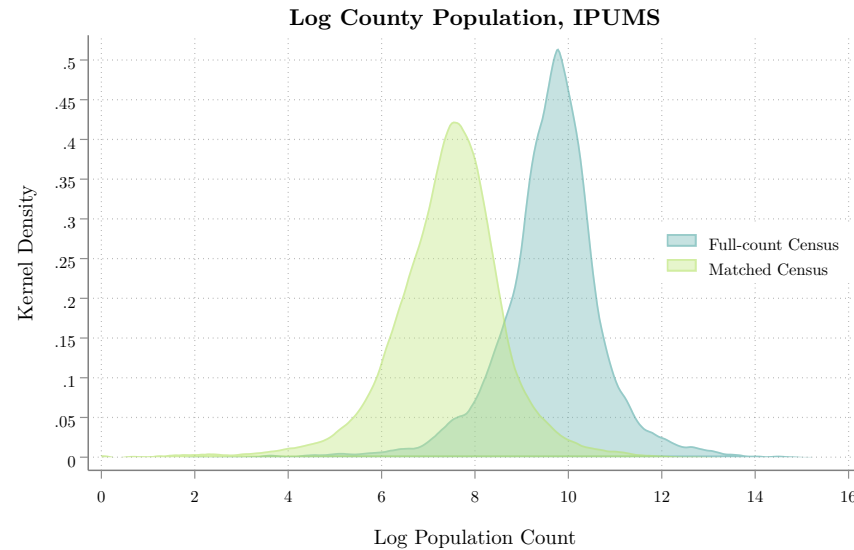


Figure A.4: Distribution of Log Population, Matched versus Full-Count census

Notes: These graphs plot county (top), city (bottom left) or non-city (bottom right) level population (in logs), comparing the distribution of population in the full-count U.S. census (1870 to 1940) to our matched version of this full-count census. The matched version is a subset of the full-count census in which individuals have been matched between two census years. We use the most conservative matching method from [Abramitzky et al. \(2020\)](#) to match individuals in adjacent census, using unique exact name and age in a 5-year age band (ABE-exact).

C. Distribution of Demographics, Full-Count vs Matched Census

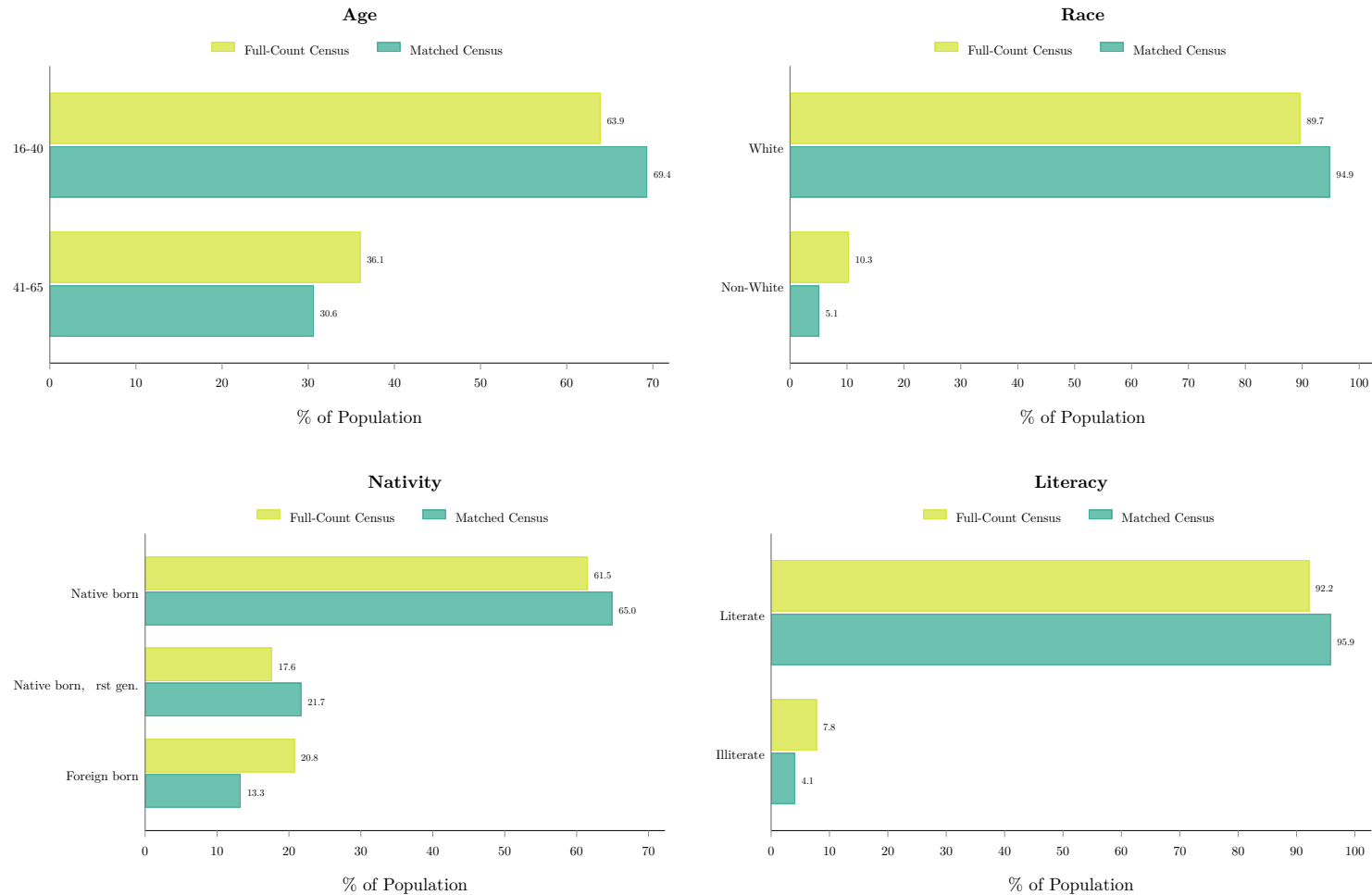


Figure A.5: Distribution of Demographics, Matched versus Full census

Notes: These graphs compare the distribution of demographics in the full-count U.S. census and our matched version of the full-count census. The matched version is a subset of the full-count census in which individuals have been matched between two census years. We use the most conservative matching method from [Abramitzky et al. \(2020\)](#) to match individuals in adjacent census, using unique exact name and age in a 5-year age band (ABE-exact). The light green bars represent the full-count census, while the dark green bars represent our matched-version of the census.

D. Documenting Birthplace (Origin Locations)

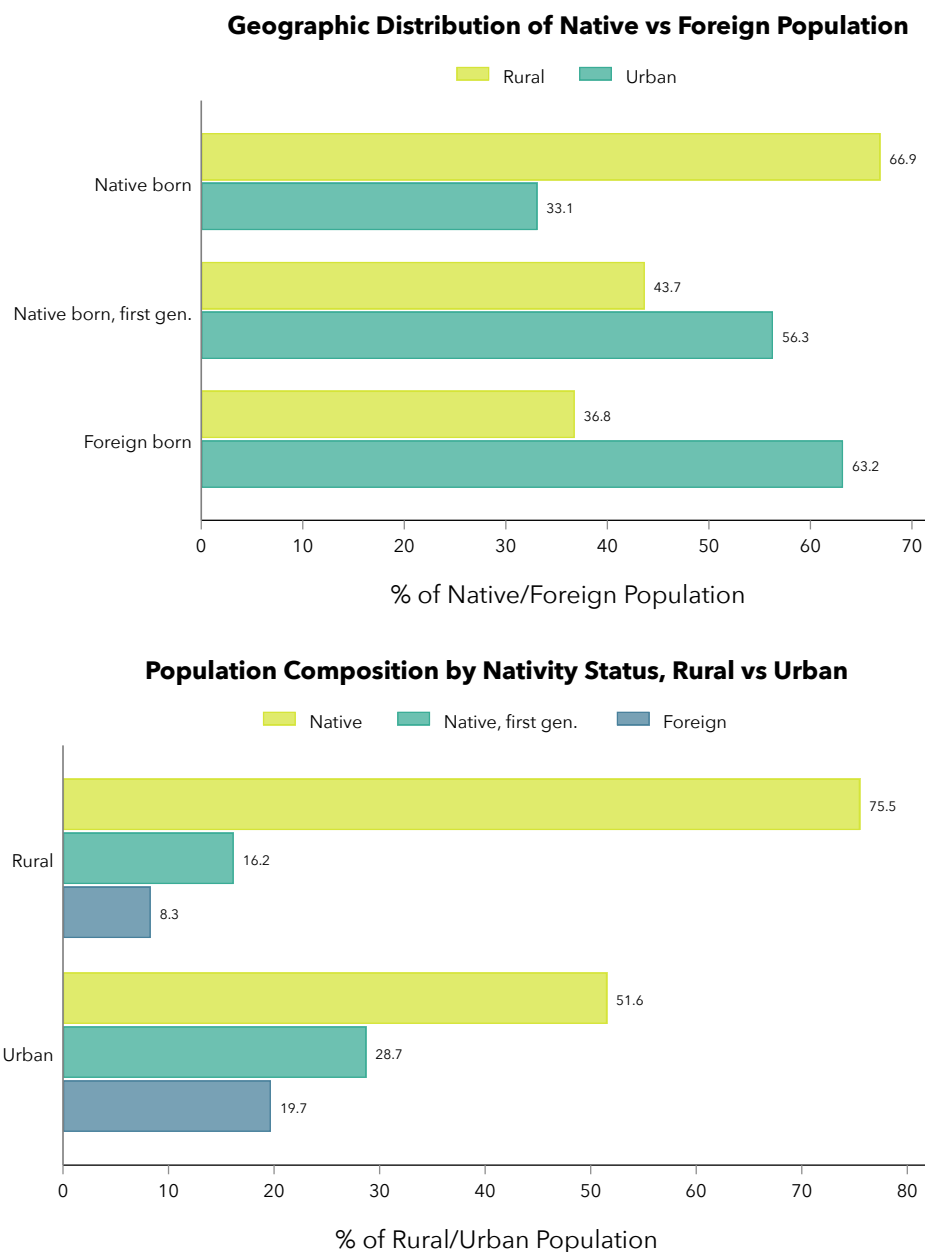


Figure A.6: Documenting Birthplace (Origin Locations)

Notes: These graphs document the birthplace of individuals in our matched census data. The top graph compares rural and urban population share across nativity status. For each nativity status (native born, native born first generation or foreign born), the total population share sums to 100. The vast majority of individuals that are native born live in rural areas (66.9%). On the contrary, only about half of native born that are first-generation U.S. citizens are living in rural places (43.7%) and most foreign-born individuals live in urban areas (62.2%). The bottom graph compares nativity status *within* a location. For each location type (rural or urban), the sum of bar plots sum to 100. 75% of individuals in rural areas are native, 16.2% are native but first-generation and 8.3% only are foreign-born. Urban areas count 51.6% of native-born, 28.7% of first-generation native born and 19.7% of foreign-born. These two graphs underline that rural areas are mostly composed of populations that immigrated in the U.S. earlier than those located in urban areas.

E. Occupations & Sectors Switching Rates

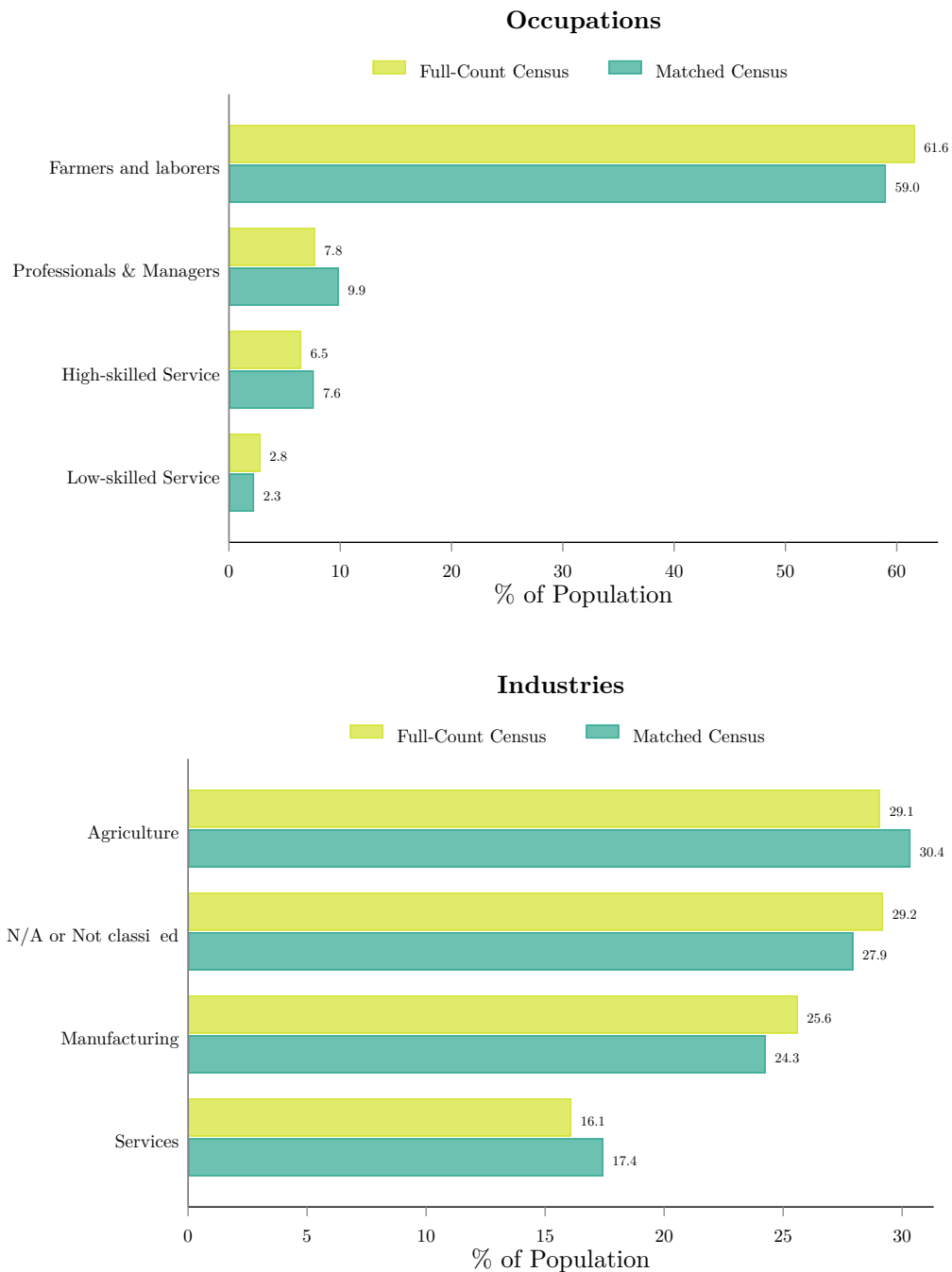


Figure A.7: Distribution of Individuals across Occupations and Industries

Notes: These graphs compare the distribution of populations by occupations and industries in the full-count U.S. census and our matched version of the full-count census. The matched version is a subset of the full-count census in which individuals have been matched between two census years. We use the most conservative matching method from [Abramitzky et al. \(2020\)](#) to match individuals in adjacent census, using unique exact name and age in a 5-year age band (ABE-exact). The light green bars represent the full-count census, while the dark green bars represent our matched-version of the census.

Table A.1: **Yearly Share of Movers Across Occupations and Sectors**

Year	Occupation 1 digit	Industry 1 Digit	Industry 3 Groups	Occupation 4 Groups	Occupation-Industry 1×1 Digit	Occupation-Industry 3× 4 Groups
1880	0.544	0.439	0.401	0.127	0.576	0.309
1900	0.636	0.531	0.489	0.146	0.667	0.322
1910	0.610	0.525	0.486	0.118	0.658	0.264
1920	0.600	0.524	0.477	0.110	0.662	0.244
1930	0.612	0.555	0.500	0.122	0.696	0.266
1940	0.623	0.603	0.511	0.162	0.728	0.326

Notes: This table details households' switching rates between intercensal years in terms of occupation and industry. The number for the first row denotes the share of people switching occupation or industry between 1870 and 1880 in the matched census data. Note that there is no 1890 full-count U.S. census. For robustness, we display switching rates for different definitions of occupation and industries. The first column refers to 1-digit occupation categories as defined by the IPUMS full-count census. The second column does the same for 1-digit industry. The two following columns aggregate industry and occupations into 3 or 4 groups. The two last columns look at switching rates across occupation-industry bins.

Table A.2: **Aggregate Share of Movers Across Occupations and Sectors**

Category	Value
Occupation 1 digit	0.611
Industry 1 Digit	0.552
Industry 3 Groups	0.491
Occupation 4 Groups	0.133
Occupation-Industry 1×1 Digit	0.686
Occupation-Industry 3× 4 Groups	0.286

Notes: This table details households' switching rates in terms of occupation and industry. Movers are defined as individuals moving at least once over the 1870-1940 period. For robustness, we display switching rates for different definitions of occupation and industries. The first row refers to 1-digit occupation categories as defined by the IPUMS full-count census. The second row does the same for 1-digit industry. The two subsequent rows aggregate industry and occupations into 3 or 4 groups. The two last rows look at switching rates across occupation-industry bins.

Table A.3: **Aggregate Share of Movers Across Geography Units and Industries**

Geography	Share Movers	Industry/Sector Def.	Share Movers
City	0.335	Occupation Groups	0.133
Urban	0.378	Sector Groups	0.391
County	0.323	Occupations × Sectors	0.286

Notes: These tables display the share of movers in out matched census data across geography units, across industry sectors and across occupations. Industries and sectors are not aggregated into groups. Movers are defined as individuals moving at least once over the 1870-1940 period.

B. Public Health Investments and Urbanization

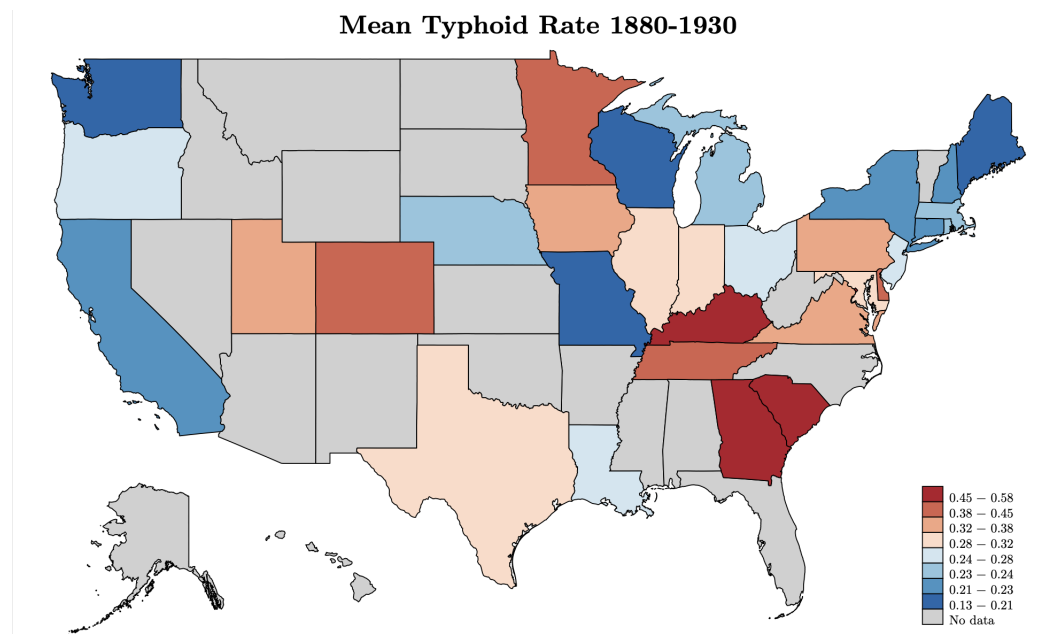
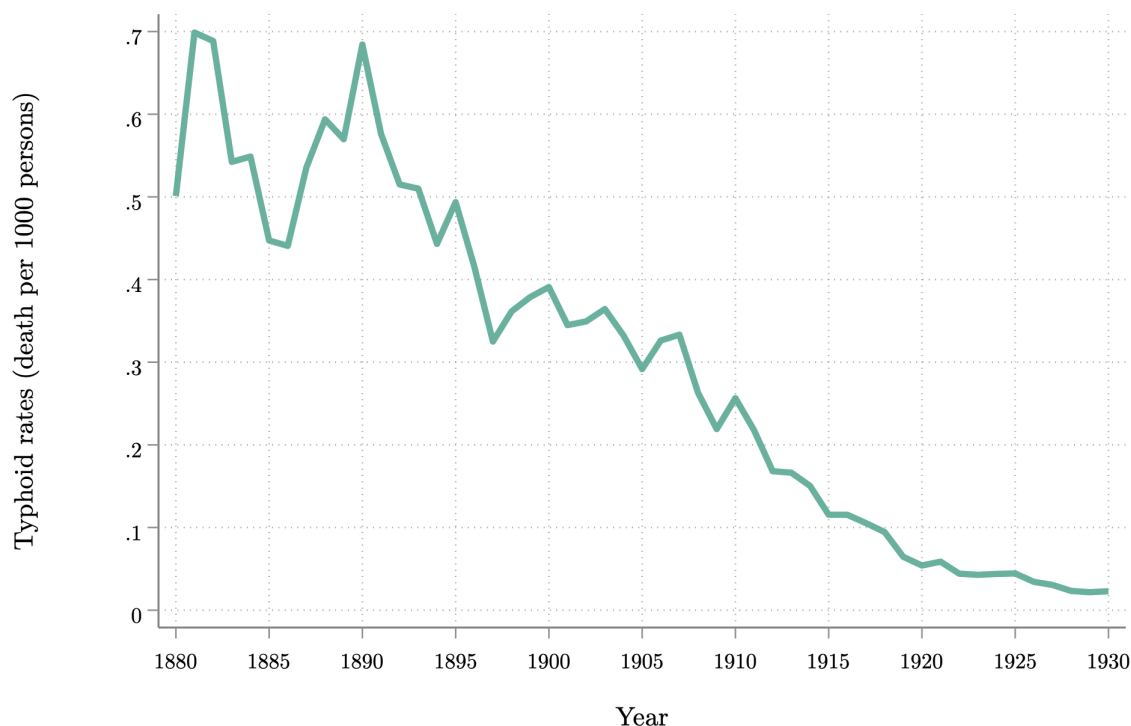


Figure B.1: Typhoid death rates over time and space

Notes: Evolution of typhoid death rates over time and space. The map plots the average typhoid rates per state over all years in the sample (1880-1930). The typhoid rates are defined as deaths per 1000 persons. The measure of typhoid rates is a consistent and balanced series (1880-1930) that combines information from Whipple with the mortality statistics. The data were obtained from Brian Beach.

C. Newspaper Coverage of Diseases

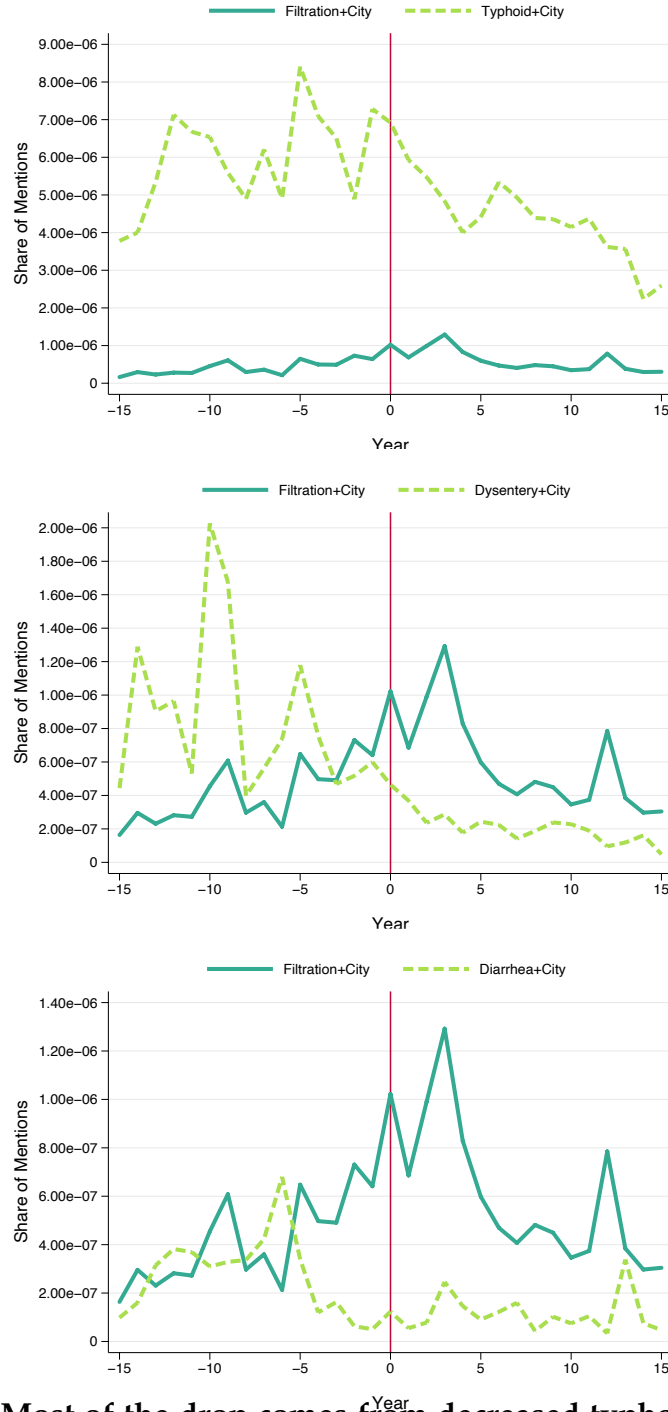


Figure C.1: **Most of the drop comes from decreased typhoid reporting**

Notes: These graphs report the share of newspaper articles mentioning waterborne diseases that are associated to a city in our dataset. The sample is currently limited to the 74 cities in our public health investment dataset, some of these cities not having an implemented water filtration system over the period. We compute the number of keyword hits having both a city name and either typhoid, dysentery or diarrhea sentences appearing next to each other in an article. Each graph is centered around the year of implementation of a water filtration system in the city. In each graph, we superpose the share of mentions to waterborne diseases (including typhoid, dysentery, and diarrhea) for these cities. The timing of water filtration investments correspond with a decrease in reports of diseases outbreaks in the newspapers, and this decrease is mostly driven by a decrease in typhoid reporting.

D. In-Migrations - Additional Empirical Results

A. Summary of migration rates

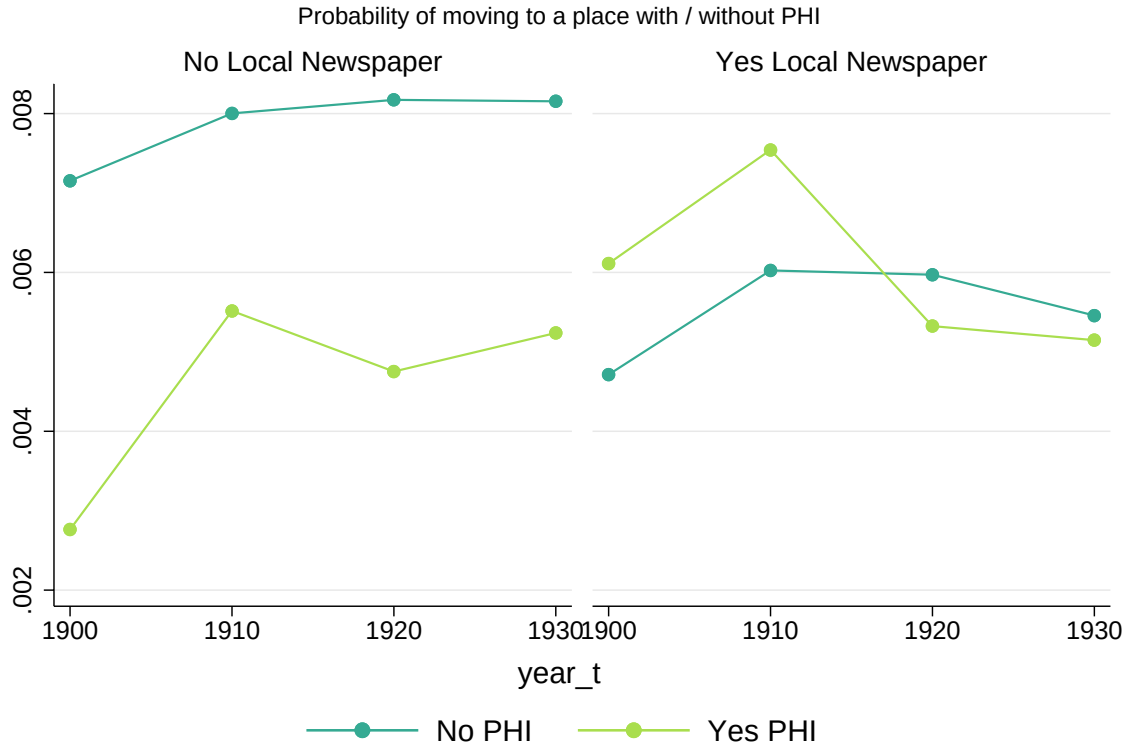


Figure D.1: Differential migration to destinations with water filtration, by whether origin has a local newspaper

Notes: This figure depicts the average migration rates across two types of origins and destinations respectively. The left panel depicts average rates from origins without a local newspaper and the right panel depicts that from origins with a local newspaper. Within each panel, the dark green line depicts average rate from that origin type to destinations without a water filtration plant and the light green line depicts that to destinations with a water filtration plant. The y-axis denotes the share of households from an origin type who migrated to a destination type for each census year. The x-axis denotes the census years.

B. In-Migrations: Typhoid rates vs PHI

In all regressions below, for the Typhoid version, the set of destination cities is restricted to the 73 cities for which we have typhoid rates, from Brian Beach's dataset (other cities have missing values for Typhoid). We do not consider individuals who did not move. The y variable is the logit log share (i.e. log share of people moving from O to D minus log share of people staying put - the outside option).

B.1. All origins, city destinations Only

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-9694*** [.3118]	-.5241* [.2753]			-5352*** [.1712]	-.4197** [.206]		
Typhoid rate (deaths by thousand)	-.7254*** [.1591]	.1174 [.2843]						
Typhoid rate (deaths by thousand)								
News Coverage $\times$ Water Filtration			.4144*** [.1452]	.39*** [.1276]			.2398*** [.07764]	.2256*** [.08223]
Water Filtration			.904*** [.07147]	.7123*** [.06052]				
Water Filtration								
Log Distance	-.8374*** [.0853]	-.8717*** [.08351]	-.624*** [.0651]	-.7582*** [.05496]	-1.044*** [.02535]	-1.033*** [.03006]	-.9329*** [.01239]	-.939*** [.01419]
Mean wage of group g in d	1.6e-07*** [3.2e-08]	7.5e-08*** [2.1e-08]	2.3e-07*** [4.0e-08]	1.8e-07*** [3.4e-08]	5.0e-08*** [9.7e-09]	4.1e-08*** [1.2e-08]	5.3e-08*** [6.4e-09]	4.1e-08*** [7.2e-09]
Log Distance $\times$ Water Filtration	-.07372 [.07525]	-.08288 [.05857]	-.2946*** [.03454]	-.2871*** [.02859]	-.1732*** [.02309]	-.1776*** [.02645]	-.2255*** [.01682]	-.224*** [.01874]
Log Transportation Cost - Donaldson Hornbeck	.6309*** [.2142]	.5248** [.2374]	.5419*** [.165]	.6776*** [.1607]	.7943*** [.05643]	.7538*** [.06611]	.8398*** [.03392]	.8347*** [.03905]
High-skilled Service	.05082*** [.00722]	.03946*** [.00715]	.1219*** [.00721]	.09633*** [.0055]	.02563*** [.00315]	.02567*** [.00361]	.03612*** [.00238]	.03802*** [.00273]
Low-skilled Service	.1349*** [.01033]	.1008*** [.0104]	.2343*** [.01685]	.1828*** [.01295]	.05073*** [.00451]	.05557*** [.00541]	.0685*** [.00386]	.07571*** [.00421]
Farmers and laborers	-.111*** [.00888]	-.05803*** [.00856]	-.1022*** [.00697]	-.06493*** [.00754]	-.02754*** [.00589]	-.02513*** [.00676]	-.0316*** [.0037]	-.02774*** [.00422]
N/A or unemployed	-.1863*** [.01474]	-.1128*** [.01018]	-.2134*** [.02181]	-.1591*** [.01702]	-.07631*** [.00673]	-.06525*** [.00711]	-.07999*** [.00493]	-.06577*** [.00516]
Share White (D-O)		-3.663*** [.811]		-3.073*** [.3903]		.06565 [.242]		-.5512 [.7449]
Share Male (D-O)		-1.955 [3.001]		-4.058*** [1.551]		3.14* [1.773]		2.756 [1.957]
Share Manuf. (D-O)		-4.246*** [1.302]		-1.008 [.6642]		-.6659 [.4997]		-1.203* [.7188]
Share Foreign Born (D-O)		6.21*** [.6206]		5.121*** [.4993]		.7869 [1.356]		1.01 [1.229]
Share Literacy (D-O)		12.77*** [2.518]		10.4*** [1.244]		3.826 [3.925]		3.866 [2.608]
Constant	-4.141*** [.2352]	-4.541*** [.2615]	-5.301*** [.182]	-5.63*** [.1788]	-4.192*** [.05596]	-4.199*** [.1342]	-5.188*** [.03938]	-5.224*** [.06666]
R2	.5994	.6681	.6341	.6849	.7986	.8001	.8092	.8103
F-Stat	369.8	381.6	804.6	630.4	866.6	453.8	1301	683.2
Dep. Var.: Mean	-4.623	-4.626	-5.152	-5.182	-4.623	-4.626	-5.152	-5.182
Dep. Var.: Std. Dev.	1.506	1.52	1.628	1.644	1.506	1.52	1.628	1.644
N Obs.	6.1e+05	4.8e+05	1.4e+06	1.1e+06	6.1e+05	4.8e+05	1.4e+06	1.1e+06

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D.1: Robustness: alternative sample keeping only O-D pairs with more than one year of observations (i.e. migration above zero)

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-.9845*** [.3242]	-.4583 [.2847]			-.5981*** [.1762]	-.4548** [.2113]		
Typhoid rate (deaths by thousand)	-.7489*** [.1653]	.1212 [.2816]						
Typhoid rate (deaths by thousand)								
News Coverage $\times$ Water Filtration			.4196*** [.1518]	.3831*** [.1395]			.182** [.0857]	.1658* [.09294]
Water Filtration			.803*** [.07059]	.6047*** [.05956]				
Water Filtration								
Log Distance	-.8163*** [.08031]	-.8764*** [.07764]	-.555*** [.06909]	-.7559*** [.05345]	-1.094*** [.02451]	-1.082*** [.02927]	-1.054*** [.01374]	-1.062*** [.01608]
Mean wage of group g in d	1.7e-07*** [3.3e-08]	7.7e-08*** [2.3e-08]	2.5e-07*** [4.0e-08]	1.7e-07*** [3.4e-08]	4.2e-08*** [1.1e-08]	3.6e-08*** [1.4e-08]	3.9e-08*** [7.6e-09]	2.9e-08*** [9.0e-09]
Log Distance $\times$ Water Filtration	-.07073 [.07806]	-.09237 [.06055]	-.3089*** [.03973]	-.3003*** [.03394]	-.1493*** [.02726]	-.1581*** [.03156]	-.141*** [.01934]	-.141*** [.02201]
Log Transportation Cost - Donaldson Hornbeck	.6173*** [.2024]	.5407** [.2266]	.4611*** [.1749]	.6627*** [.1679]	.7988*** [.05399]	.7611*** [.06393]	.8217*** [.03179]	.8177*** [.03762]
High-skilled Service	.0455*** [.00699]	.03482*** [.00684]	.1087*** [.00634]	.08504*** [.00521]	.02484*** [.0031]	.02465*** [.00356]	.03284*** [.00248]	.03497*** [.00287]
Low-skilled Service	.1358*** [.01021]	.1001*** [.01049]	.2244*** [.0131]	.1676*** [.01138]	.05066*** [.00448]	.05494*** [.00543]	.0676*** [.0039]	.07383*** [.00436]
Farmers and laborers	-.1129*** [.00843]	-.06115*** [.00916]	-.1234*** [.00676]	-.07826*** [.00832]	-.02479*** [.00586]	-.02225*** [.00674]	-.02834*** [.00425]	-.02453*** [.00489]
N/A or unemployed	-.1855*** [.01375]	-.1122*** [.0104]	-.2302*** [.01849]	-.158*** [.01395]	-.06899*** [.00645]	-.05928*** [.00695]	-.0703*** [.00484]	-.05816*** [.00516]
Share White (D-O)		-3.836*** [.8642]		-3.716*** [.4731]		-.01339 [.2622]		-.4047 [.6129]
Share Male (D-O)		-2.194 [3.053]		-4.588** [1.808]		2.685 [1.808]		2.696* [1.499]
Share Manuf. (D-O)		-4.461*** [1.374]		-1.254 [.8202]		-.6432 [.5574]		-1.08* [.6441]
Share Foreign Born (D-O)		6.551*** [.6168]		6.105*** [.4555]		.744 [1.537]		1.461 [1.572]
Share Literacy (D-O)		13.55*** [2.654]		12.57*** [1.418]		3.423 [4.564]		5.399 [4.058]
Constant	-4.153*** [.2273]	-4.599*** [.2516]	-5.184*** [.1933]	-5.594*** [.1882]	-4.154*** [.05229]	-4.164*** [.1394]	-5*** [.03684]	-5.059*** [.08387]
R2	.601	.6707	.6225	.6826	.8085	.8095	.8218	.8224
F-Stat	329.7	351	560.2	578.8	953.6	472.3	1478	741
Dep. Var.: Mean	-4.583	-4.583	-4.997	-5.01	-4.583	-4.583	-4.997	-5.01
Dep. Var.: Std. Dev.	1.519	1.533	1.668	1.683	1.519	1.533	1.668	1.683
N Obs.	5.5e+05	4.3e+05	1.1e+06	8.6e+05	5.5e+05	4.3e+05	1.1e+06	8.6e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

B.2. Rural origins and city destinations

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-5806*** [.192]	-3071 [.1897]			-3199*** [.1117]	-2805** [.1375]		
Typhoid rate (deaths by thousand)	-8369*** [.1695]	-1383 [.2316]						
Typhoid rate (deaths by thousand)								
News Coverage $\times$ Water Filtration			.2551*** [.08973]	.2531*** [.0809]			.1431*** [.0489]	.1286** [.05236]
Water Filtration			.9123*** [.07325]	.7409*** [.06707]				
Water Filtration								
Log Distance	-.9487*** [.06677]	-1.02*** [.06379]	-.6609*** [.05364]	-.8293*** [.04094]	-1.153*** [.02861]	-1.136*** [.03277]	-.975*** [.01491]	-.9838*** [.01704]
Dummy = 1 if O= D	2.935*** [.1994]	3.058*** [.2078]	3.063*** [.1566]	3.428*** [.1443]	3.483*** [.09084]	3.425*** [.108]	3.799*** [.05606]	3.833*** [.0664]
Mean wage of group g in d	9.6e-08*** [2.2e-08]	3.7e-08** [1.7e-08]	1.7e-07*** [2.9e-08]	1.4e-07*** [2.5e-08]	3.4e-08*** [8.1e-09]	2.8e-08*** [9.5e-09]	2.5e-08*** [5.7e-09]	1.5e-08** [6.6e-09]
Log Distance $\times$ Water Filtration	-.09105 [.05621]	-.1126*** [.04188]	-.3192*** [.0335]	-.3281*** [.02589]	-.2426*** [.02454]	-.2456*** [.02678]	-.3292*** [.02034]	-.33*** [.02125]
Log Transportation Cost - Donaldson Hornbeck	.744*** [.1655]	.769*** [.1745]	.5374*** [.1355]	.7662*** [.1245]	1.043*** [.06335]	.9935*** [.07208]	.9861*** [.04301]	.9939*** [.05056]
High-skilled Service	.04118*** [.00655]	.03743*** [.00651]	.1063*** [.00555]	.08814*** [.00469]	.0277*** [.00369]	.02964*** [.00404]	.03846*** [.00254]	.04185*** [.00289]
Low-skilled Service	.08758*** [.00873]	.0647*** [.00962]	.1687*** [.00938]	.1302*** [.00796]	.03237*** [.00575]	.03511*** [.00687]	.04915*** [.00449]	.05586*** [.00522]
Farmers and laborers	-.09851*** [.00844]	-.05881*** [.00704]	-.08817*** [.00669]	-.06255*** [.00572]	-.03953*** [.00495]	-.03785*** [.00576]	-.03993*** [.00297]	-.0359*** [.00339]
N/A or unemployed	-.1354*** [.01201]	-.08541*** [.00864]	-.1545*** [.01594]	-.1203*** [.01266]	-.06657*** [.00577]	-.05795*** [.00632]	-.05849*** [.00409]	-.04639*** [.00456]
Share White (D-O)		-2.719*** [.6726]		-2.27*** [.3054]		.101 [.3028]		.2272 [.3835]
Share Male (D-O)		-4.208* [2.139]		-5.034*** [1.224]		-.7747 [.6075]		.3511 [1.163]
Share Manuf. (D-O)		-2.28** [1.049]		.05326 [.5097]		.1842 [.303]		-.4418 [.4371]
Share Foreign Born (D-O)		4.961*** [.4361]		4.185*** [.4109]		-1.867*** [.6359]		-.3671 [.5021]
Share Literacy (D-O)		9.568*** [1.992]		8.298*** [.9352]		-5.352** [2.261]		.9034 [1.747]
Constant	-4.052*** [.1912]	-4.648*** [.2217]	-4.956*** [.1592]	-5.492*** [.1485]	-4.335*** [.06546]	-4.006*** [.1187]	-5.116*** [.05147]	-5.085*** [.07239]
R2	.6708	.7071	.6686	.7014	.8037	.8032	.8069	.8046
F-Stat	338.2	318.7	1222	912.9	879.6	512.3	3110	1567
Dep. Var.: Mean	-4.421	-4.412	-4.7	-4.724	-4.421	-4.412	-4.7	-4.724
Dep. Var.: Std. Dev.	1.462	1.472	1.54	1.543	1.462	1.472	1.54	1.543
N Obs.	3.8e+05	3.0e+05	9.5e+05	7.6e+05	3.8e+05	3.0e+05	9.5e+05	7.6e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D.2: Robustness: alternative sample keeping only O-D pairs with more than one year of observations (i.e. migration shares above zero)

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-.5665*** [.1988]	-.2352 [.1986]			-.3354*** [.1101]	-.2803** [.1324]		
Typhoid rate (deaths by thousand)	-.8764*** [.1793]	-.1614 [.2321]						
Typhoid rate (deaths by thousand)								
News Coverage $\times$ Water Filtration			.2458*** [.09051]	.2424*** [.08391]			.09742* [.05169]	.08406 [.05654]
Water Filtration			.8106*** [.071]	.6222*** [.0615]				
Water Filtration								
Log Distance	-.9345*** [.06292]	-1.03*** [.05846]	-.5925*** [.05675]	-.829*** [.03837]	-1.198*** [.02795]	-1.18*** [.0324]	-1.069*** [.0149]	-1.08*** [.01731]
Dummy = 1 if O= D	2.932*** [.1963]	3.089*** [.2025]	2.914*** [.1663]	3.4*** [.151]	3.516*** [.08852]	3.456*** [.1065]	3.815*** [.05245]	3.856*** [.06323]
Mean wage of group g in d	9.9e-08*** [2.2e-08]	3.8e-08** [1.8e-08]	1.8e-07*** [2.9e-08]	1.3e-07*** [2.5e-08]	2.8e-08*** [9.6e-09]	2.4e-08** [1.2e-08]	1.7e-08** [7.2e-09]	8.5e-09 [8.6e-09]
Log Distance $\times$ Water Filtration	-.0918 [.05842]	-.1285*** [.04311]	-.3416*** [.03975]	-.3506*** [.02994]	-.2293*** [.02464]	-.235*** [.02763]	-.2641*** [.0197]	-.2667*** [.02174]
Log Transportation Cost - Donaldson Hornbeck	.7436*** [.1559]	.8056*** [.1631]	.4485*** [.1426]	.7543*** [.1264]	1.063*** [.06296]	1.016*** [.07302]	.9394*** [.04203]	.9514*** [.05052]
High-skilled Service	.03397*** [.00643]	.03002*** [.00635]	.08627*** [.00514]	.07105*** [.00486]	.02623*** [.00367]	.02681*** [.00405]	.03246*** [.00265]	.03525*** [.00303]
Low-skilled Service	.08509*** [.00904]	.06164*** [.00992]	.15*** [.00711]	.1106*** [.00806]	.03406*** [.00569]	.03533*** [.00678]	.0474*** [.0044]	.05141*** [.0052]
Farmers and laborers	-.1022*** [.00811]	-.0635*** [.00787]	-.1127*** [.00628]	-.08012*** [.00617]	-.03742*** [.00505]	-.0361*** [.00593]	-.04233*** [.00329]	-.03882*** [.00378]
N/A or unemployed	-.1361*** [.01098]	-.08743*** [.00882]	-.1731*** [.01284]	-.126*** [.01011]	-.06159*** [.00558]	-.05444*** [.00627]	-.05795*** [.00395]	-.0485*** [.00447]
Share White (D-O)		-2.847*** [.7323]		-2.749*** [.3808]		.00378 [.3385]		.3458 [.4141]
Share Male (D-O)		-4.627** [2.159]		-5.871*** [1.412]		-1.347** [.5908]		-.3396 [1.344]
Share Manuf. (D-O)		-2.369** [1.148]		.1059 [.6608]		.1848 [.328]		-.1883 [.4593]
Share Foreign Born (D-O)		5.257*** [.4356]		5.02*** [.3547]		-1.892*** [.6383]		-.1001 [.6948]
Share Literacy (D-O)		10.21*** [2.177]		10.08*** [1.111]		-5.829** [2.298]		1.245 [2.56]
Constant	-4.068*** [.1845]	-4.73*** [.2169]	-4.79*** [.1674]	-5.472*** [.1545]	-4.327*** [.06364]	-4.008*** [.1169]	-4.92*** [.04938]	-4.914*** [.08162]
R2	.6707	.7071	.6515	.6905	.8113	.8102	.8127	.8096
F-Stat	301.2	286	903.3	774.6	880.8	468.6	2601	1278
Dep. Var.: Mean	-4.348	-4.337	-4.458	-4.467	-4.348	-4.337	-4.458	-4.467
Dep. Var.: Std. Dev.	1.468	1.477	1.552	1.553	1.468	1.477	1.552	1.553
N Obs.	3.4e+05	2.7e+05	7.2e+05	5.7e+05	3.4e+05	2.7e+05	7.2e+05	5.7e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

C. In-Migrations: Typhoid rates vs PHI, limiting the sample of origins to only towns with newspapers

All regressions below are the same as above, but this time we limit origins to towns with newspapers only.

C.1. All origins, city destinations Only

FEs:	Dependent Var.: Probability of moving							
	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-1.283*** [.4321]	-.06172 [.3512]			-.7668*** [.2735]	-.4549* [.2603]		
Typhoid rate (deaths by thousand)	-.4293* [.2335]	-.09899 [.307]						
News Coverage $\times$ Water Filtration			.65** [.2904]	.6944*** [.2161]			.4444*** [.1578]	.3581** [.1491]
Water Filtration			.6702*** [.1931]	.3636*** [.1194]				
Log Distance	-.8238*** [.1043]	-.8686*** [.09894]	-.6164*** [.07919]	-.7361*** [.06694]	-1.028*** [.02637]	-1.016*** [.03157]	-.9253*** [.01435]	-.9306*** [.01656]
Mean wage of group g in d	2.0e-07*** [3.9e-08]	8.7e-08*** [2.6e-08]	2.7e-07*** [4.8e-08]	2.1e-07*** [4.0e-08]	5.4e-08*** [1.3e-08]	4.6e-08*** [1.5e-08]	6.2e-08*** [7.7e-09]	5.2e-08*** [8.8e-09]
Log Distance $\times$ Water Filtration	-.06374 [.08663]	-.06173 [.06835]	-.2952*** [.03719]	-.2701*** [.0335]	-.1113*** [.02764]	-.1125*** [.03138]	-.1668*** [.02148]	-.1624*** [.02414]
Log Transportation Cost - Donaldson Hornbeck	.6119** [.256]	.4868* [.2762]	.5505*** [.1952]	.6332*** [.1871]	.7022*** [.05867]	.6561*** [.06896]	.778*** [.03786]	.7693*** [.04385]
Share White (D-O)		-4.316*** [.9655]		-3.696*** [.5019]		.03662 [.2662]		-.08264 [.4434]
Share Male (D-O)		.3063 [3.762]		-3.024 [1.869]		3.404 [2.338]		5.68** [2.252]
Share Manuf. (D-O)		-5.366*** [1.444]		-1.549* [.8049]		-.5133 [.4642]		-.7896 [.5183]
Share Foreign Born (D-O)		7.273*** [.7045]		5.936*** [.5874]		.2411 [1.035]		.2048 [.7616]
Share Literacy (D-O)		15.76*** [3.026]		12.78*** [1.639]		2.782 [3.191]		3.943 [2.482]
Occupation FE	✓	✓	✓	✓	✓	✓	✓	✓
Constant	-4.227*** [.2678]	-4.59*** [.2657]	-5.548*** [.2022]	-5.736*** [.1896]	-4.155*** [.05777]	-4.137*** [.0843]	-5.346*** [.04683]	-5.348*** [.05768]
R2	.5591	.6572	.6008	.6647	.8066	.8087	.8135	.8149
F-Stat	315	336.1	616.9	491.9	813.4	490.4	1085	667.8
Dep. Var.: Mean	-4.76	-4.77	-5.381	-5.42	-4.76	-4.77	-5.381	-5.42
Dep. Var.: Std. Dev.	1.522	1.536	1.642	1.658	1.522	1.536	1.642	1.658
N Obs.	4.1e+05	3.2e+05	9.6e+05	7.7e+05	4.1e+05	3.2e+05	9.6e+05	7.7e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D.3: Robustness: alternative sample keeping only O-D pairs with more than one year of observations (i.e. migration shares above zero)

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-1.331*** [.4691]	-.06617 [.3772]			-.9285*** [.3159]	-.4373 [.2678]		
Typhoid rate (deaths by thousand)	-.4177 [.2822]	-.04742 [.2489]						
News Coverage $\times$ Water Filtration			.646* [.3671]	.7689*** [.2678]			.4984** [.201]	.3595* [.1962]
Water Filtration			.576** [.2883]	.167 [.1823]				
Log Distance	-.7904*** [.09715]	-.8636*** [.09146]	-.5244*** [.08389]	-.7172*** [.06535]	-1.066*** [.02515]	-1.054*** [.03014]	-1.03*** [.01571]	-1.038*** [.01839]
Mean wage of group g in d	2.1e-07*** [4.2e-08]	9.5e-08*** [3.0e-08]	3.0e-07*** [4.9e-08]	2.0e-07*** [4.1e-08]	4.9e-08*** [1.4e-08]	4.5e-08*** [1.7e-08]	4.8e-08*** [8.9e-09]	3.9e-08*** [1.1e-08]
Log Distance $\times$ Water Filtration	-.05731 [.08894]	-.06984 [.07013]	-.3081*** [.04263]	-.2795*** [.03954]	-.08669*** [.03269]	-.09186*** [.0374]	-.0841*** [.02428]	-.07979*** [.02754]
Log Transportation Cost - Donaldson Hornbeck	.5813** [.2394]	.4919* [.2624]	.4405** [.2065]	.5914*** [.1957]	.6948*** [.05533]	.6532*** [.06528]	.7433*** [.03599]	.7332*** [.04241]
High-skilled Service	.04755*** [.00796]	.03004*** [.00805]	.1183*** [.00743]	.08965*** [.00619]	.02025*** [.00352]	.01829*** [.00402]	.02926*** [.00284]	.03004*** [.00324]
Low-skilled Service	.1691*** [.01312]	.1144*** [.01355]	.2701*** [.01683]	.1994*** [.01482]	.05751*** [.00533]	.06084*** [.00641]	.07484*** [.00437]	.08058*** [.00511]
Farmers and laborers	-.1275*** [.00968]	-.0666*** [.01062]	-.136*** [.00809]	-.08056*** [.01018]	-.02429*** [.00632]	-.02212*** [.00728]	-.02794*** [.00469]	-.02493*** [.00527]
N/A or unemployed	-.2134*** [.01556]	-.1238*** [.0128]	-.2614*** [.02184]	-.1739*** [.01658]	-.07131*** [.00733]	-.06144*** [.0079]	-.07554*** [.00542]	-.06364*** [.00568]
Share White (D-O)		-4.441*** [1.02]		-4.384*** [.605]		-.04987 [.2551]		.2002 [.4325]
Share Male (D-O)		-.07077 [3.772]		-3.696* [2.132]		2.816 [2.296]		4.656** [2.16]
Share Manuf. (D-O)		-5.548*** [1.501]		-1.829* [.9764]		-.4813 [.517]		-.7627 [.5223]
Share Foreign Born (D-O)		7.565*** [.689]		7.044*** [.5299]		.08861 [1.215]		1.166 [1.124]
Share Literacy (D-O)		16.03*** [3.1]		15.14*** [1.788]		1.968 [3.85]		6.934* [3.716]
Constant	-4.232*** [.2582]	-4.641*** [.2532]	-5.413*** [.2152]	-5.658*** [.1978]	-4.098*** [.05597]	-4.1*** [.08858]	-5.16*** [.05539]	-5.171*** [.06236]
R2	.5601	.6593	.5885	.6653	.8148	.8167	.8265	.8279
F-Stat	277.2	311.5	428.3	419.6	881.9	496.6	1254	729.4
Dep. Var.: Mean	-4.725	-4.734	-5.227	-5.25	-4.725	-4.734	-5.227	-5.25
Dep. Var.: Std. Dev.	1.532	1.546	1.677	1.693	1.532	1.546	1.677	1.693
N Obs.	3.7e+05	2.9e+05	7.2e+05	5.7e+05	3.7e+05	2.9e+05	7.2e+05	5.7e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

C.2. Rural origins and city destinations

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-.7621*** [.2794]	.1269 [.297]			-.3358** [.1532]	-.2537* [.1444]		
Typhoid rate (deaths by thousand)	-.7444*** [.2588]	-.3641 [.3051]						
News Coverage $\times$ Water Filtration			.2961 [.2181]	.2961** [.1354]			.1613* [.08879]	.06715 [.09191]
Water Filtration			.9244*** [.1774]	.6863*** [.08818]				
Log Distance	-.9775*** [.08601]	-1.065*** [.07692]	-.6889*** [.07081]	-.8581*** [.05468]	-1.208*** [.03175]	-1.188*** [.03728]	-1.048*** [.01897]	-1.053*** [.02177]
Mean wage of group g in d	1.2e-07*** [3.1e-08]	3.7e-08 [2.2e-08]	2.1e-07*** [4.0e-08]	1.7e-07*** [3.4e-08]	3.9e-08*** [1.2e-08]	3.3e-08*** [1.4e-08]	3.1e-08*** [7.7e-09]	2.2e-08** [9.0e-09]
Log Distance $\times$ Water Filtration	-.08755 [.06673]	-.09477* [.05201]	-.3466*** [.03623]	-.3334*** [.03101]	-.1804*** [.02533]	-.1765*** [.02777]	-.2857*** [.02035]	-.2782*** [.02159]
Log Transportation Cost - Donaldson Hornbeck	.7967*** [.2088]	.8084*** [.2082]	.6015*** [.1748]	.8055*** [.1575]	1.085*** [.06891]	1.028*** [.0792]	1.083*** [.05373]	1.076*** [.062]
High-skilled Service	.04406*** [.00868]	.0306*** [.00894]	.1277*** [.00791]	.1021*** [.00678]	.02271*** [.00495]	.02062*** [.00551]	.03546*** [.00356]	.03735*** [.00408]
Low-skilled Service	.1129*** [.01384]	.07571*** [.01391]	.2066*** [.01374]	.1597*** [.01199]	.03997*** [.00798]	.03815*** [.00956]	.05233*** [.00623]	.05636*** [.00747]
Farmers and laborers	-.1206*** [.01117]	-.06638*** [.00919]	-.09991*** [.00908]	-.06228*** [.0077]	-.03837*** [.00621]	-.03859*** [.00739]	-.03743*** [.0037]	-.03479*** [.00422]
N/A or unemployed	-.1615*** [.01508]	-.09556*** [.01163]	-.1752*** [.02161]	-.1303*** [.01704]	-.06702*** [.00736]	-.06027*** [.00842]	-.05617*** [.00506]	-.04613*** [.00567]
Share White (D-O)		-3.64*** [.8779]		-3.071*** [.4445]		-1.807 [.3834]		-.7861 [.7865]
Share Male (D-O)		-2.276 [2.959]		-4.384*** [1.666]		-.9168 [.9116]		3.377 [2.063]
Share Manuf. (D-O)		-3.6*** [1.234]		-.4991 [.6584]		.2869 [.3861]		-1.604* [.8732]
Share Foreign Born (D-O)		6.181*** [.5318]		5.178*** [.5032]		-2.553*** [.5762]		.8324 [1.095]
Share Literacy (D-O)		13.22*** [2.664]		11.11*** [1.414]		-7.968*** [2.143]		4.171 [3.73]
Constant	-4.243*** [.2232]	-4.788*** [.2247]	-5.295*** [.19]	-5.723*** [.1697]	-4.465*** [.06502]	-4.137*** [.09459]	-5.389*** [.06035]	-5.365*** [.08706]
R2	.6079	.6749	.5639	.6258	.7968	.7982	.7789	.7803
F-Stat	235.9	253	536.1	469.3	619.9	367.4	961.3	537.7
Dep. Var.: Mean	-4.665	-4.649	-5.112	-5.122	-4.665	-4.649	-5.112	-5.122
Dep. Var.: Std. Dev.	1.437	1.454	1.429	1.445	1.437	1.454	1.429	1.445
N Obs.	1.9e+05	1.5e+05	4.6e+05	3.6e+05	1.9e+05	1.5e+05	4.6e+05	3.6e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D.4: Robustness: alternative sample keeping only O-D pairs with more than one year of observations (i.e. migration shares > 0)

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-.7758** [.3029]	.1582 [.3204]			-.4329*** [.1651]	-.2515* [.1427]		
Typhoid rate (deaths by thousand)	-.753** [.3011]	-.3695 [.2859]						
News Coverage $\times$ Water Filtration			.2439 [.3108]	.2825 [.1895]			.2178** [.1085]	.07402 [.1319]
Water Filtration			.8601*** [.2807]	.5517*** [.1596]				
Log Distance	-.9394*** [.07748]	-1.055*** [.06752]	-.5852*** [.07471]	-.8415*** [.05094]	-1.239*** [.03054]	-1.22*** [.03599]	-1.147*** [.019]	-1.157*** [.02231]
Mean wage of group g in d	1.3e-07*** [3.1e-08]	4.2e-08* [2.5e-08]	2.4e-07*** [4.0e-08]	1.7e-07*** [3.4e-08]	3.5e-08** [1.4e-08]	3.4e-08** [1.7e-08]	2.4e-08** [9.6e-09]	1.8e-08 [1.2e-08]
Log Distance $\times$ Water Filtration	-.08406 [.06733]	-.1085** [.05205]	-.3684*** [.04253]	-.3491*** [.0361]	-.1537*** [.02655]	-.1526*** [.02958]	-.1895*** [.02164]	-.1815*** [.02382]
Log Transportation Cost - Donaldson Hornbeck	.7691*** [.189]	.8247*** [.1878]	.473** [.1844]	.7764*** [.1586]	1.083*** [.06707]	1.034*** [.0783]	1.036*** [.05195]	1.037*** [.06126]
High-skilled Service	.03533*** [.0083]	.02162** [.00852]	.1048*** [.00716]	.08122*** [.00701]	.02119*** [.00499]	.01806*** [.00544]	.02884*** [.00373]	.02915*** [.0042]
Low-skilled Service	.1093*** [.01438]	.07086*** [.01427]	.1855*** [.01159]	.1321*** [.01246]	.03972*** [.00779]	.03718*** [.00929]	.04613*** [.00617]	.04707*** [.00746]
Farmers and laborers	-.1269*** [.01031]	-.07425*** [.01056]	-.1307*** [.00875]	-.08358*** [.00884]	-.03892*** [.00658]	-.03934*** [.00793]	-.04165*** [.00438]	-.04016*** [.00511]
N/A or unemployed	-.1655*** [.0134]	-.1008*** [.01217]	-.2028*** [.01739]	-.139*** [.01381]	-.06486*** [.00721]	-.05912*** [.00841]	-.05948*** [.00509]	-.05257*** [.00586]
Share White (D-O)		-3.774*** [.9528]		-3.753*** [.571]		-.1974 [.4139]		-.2184 [.8701]
Share Male (D-O)		-2.705 [2.907]		-5.393*** [1.902]		-1.351 [.9611]		3.416 [2.775]
Share Manuf. (D-O)		-3.697*** [1.319]		-.5063 [.8522]		.2957 [.4249]		-1.518 [1.046]
Share Foreign Born (D-O)		6.419*** [.5196]		6.247*** [.4182]		-2.606*** [.6018]		2.123 [1.815]
Share Literacy (D-O)		13.36*** [2.814]		13.63*** [1.6]		-8.559*** [2.305]		8.232 [6.066]
Constant	-4.248*** [.2108]	-4.857*** [.2102]	-5.095*** [.201]	-5.673*** [.1729]	-4.436*** [.06236]	-4.137*** [.09702]	-5.174*** [.05933]	-5.22*** [.1078]
R2	.6047	.6727	.5342	.611	.8016	.8025	.7834	.7845
F-Stat	199.6	238.4	318.6	402.5	633	348.9	970.7	518.4
Dep. Var.: Mean	-4.594	-4.578	-4.886	-4.88	-4.594	-4.578	-4.886	-4.88
Dep. Var.: Std. Dev.	1.432	1.446	1.428	1.444	1.432	1.446	1.428	1.444
N Obs.	1.7e+05	1.3e+05	3.3e+05	2.6e+05	1.7e+05	1.3e+05	3.3e+05	2.6e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

E. In-Migrations: Placebo Tests

This section presents additional in-migration regression results based on placebo tests that use past migration flows.

A. All origins, city destinations Only

Table E.1: Placebo test, dependent variable: migrations 10 years ago

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-.4901 [.5222]	.1448 [.436]			.3786 [.3567]	.39 [.3654]		
Typhoid rate (deaths by thousand)	.2348 [.9182]	.4518 [1.065]						
News Coverage $\times$ Water Filtration			-.1393 [.1931]	.02432 [.187]			-.3805** [.1714]	-.3933** [.1892]
Water Filtration			-.1792 [.162]	-.2596 [.1949]				
Log Distance	-.5007*** [.08821]	-.4953*** [.09394]	-.08921 [.08694]	-.01018 [.09205]	-.5663*** [.06493]	-.499*** [.07079]	.121** [.05248]	.1035* [.05757]
Mean wage of group g in d	1.6e-07*** [4.6e-08]	1.2e-07*** [4.2e-08]	-1.1e-07 [6.7e-08]	-5.4e-08 [7.3e-08]	1.2e-07*** [2.7e-08]	1.1e-07*** [3.1e-08]	1.4e-07*** [1.8e-08]	1.4e-07*** [2.0e-08]
Log Distance $\times$ Water Filtration	.05712 [.05476]	.06356 [.05121]	-.259*** [.06719]	-.202*** [.06869]	-.2261*** [.06739]	-.2128*** [.07064]	-.5676*** [.07045]	-.5458*** [.07311]
Log Transportation Cost - Donaldson Hornbeck	.5993*** [.1977]	.493** [.2141]	.5387** [.2181]	.3224 [.2244]	.9411*** [.1459]	.784*** [.1655]	.6988*** [.1301]	.7181*** [.1467]
High-skilled Service	-.02243 [.01598]	-.0333* [.01866]	-.1666*** [.01637]	-.1457*** [.01692]	-.01628 [.0124]	-.01254 [.01286]	-.02028* [.01042]	-.02165* [.01136]
Low-skilled Service	.1016*** [.02423]	.08104*** [.02763]	-.1631*** [.02944]	-.1325*** [.03184]	.05188** [.02615]	.08181*** [.02021]	.06389*** [.01964]	.08047*** [.01722]
Farmers and laborers	-.0643*** [.01491]	-.03447** [.01608]	-.02287 [.01857]	-.03912** [.01965]	-.02214 [.01412]	-.02135 [.01421]	-.05419*** [.01091]	-.05454*** [.01224]
N/A or unemployed	-.1327*** [.01711]	-.09226*** [.0145]	.02619 [.0316]	.00035 [.03167]	-.08505*** [.01393]	-.07525*** [.01357]	-.1229*** [.01188]	-.1107*** [.01295]
Share White (D-O)		.09846 [1.001]		3.614*** [.7379]		-20.52 [12.64]		-25.79 [16.4]
Share Male (D-O)		3.987 [5.083]		10.39*** [2.838]		58.39*** [20.2]		25.02 [21.22]
Share Manuf. (D-O)		-5.267*** [1.796]		-3.601*** [.9265]		-36.16** [14.12]		-31.19* [17.6]
Share Foreign Born (D-O)		2.319*** [.8872]		-2.341*** [.7613]		55.55** [21.5]		35.85** [18.14]
Share Literacy (D-O)		-3.159 [2.781]		-15.17*** [2.27]		223.9*** [80.16]		135.8** [62.92]
Constant	-4.321*** [.2379]	-4.442*** [.2573]	-4.107*** [.211]	-3.975*** [.22]	-4.674*** [.1565]	-7.516*** [1.144]	-4.582*** [.1406]	-5.303*** [.3532]
R2	.3495	.3517	.2479	.2528	.4206	.4297	.4662	.4762
F-Stat	23.26	18.93	21.55	20.6	34.88	25.54	41.99	28.97
Dep. Var.: Mean	-4.178	-4.311	-3.622	-3.717	-4.178	-4.311	-3.622	-3.717
Dep. Var.: Std. Dev.	2.771	2.784	3.52	3.561	2.771	2.784	3.52	3.561
N Obs.	2.8e+05	2.2e+05	6.3e+05	5.1e+05	2.8e+05	2.2e+05	6.3e+05	5.1e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table E.2: Placebo test, dependent variable: migrations 20 years ago

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	.2003 [.6155]	.9623*** [.348]			1.213*** [.3111]	1.273*** [.3225]		
Typhoid rate (deaths by thousand)	-.1384 [.7322]	.1276 [.7467]						
News Coverage $\times$ Water Filtration			.16 [.2514]	.1145 [.2717]			-.1444 [.1676]	-.3351** [.159]
Water Filtration			-.8935*** [.2348]	-.726*** [.2801]				
Log Distance	-.06423 [.1165]	.109 [.1132]	.06194 [.103]	.3368*** [.1084]	-.2636*** [.08821]	-.1705 [.1067]	.2072*** [.05208]	.2531*** [.06168]
Mean wage of group g in d	2.2e-07*** [7.2e-08]	1.2e-07** [5.0e-08]	-2.1e-07*** [8.1e-08]	-1.6e-07* [8.2e-08]	1.1e-07*** [3.4e-08]	9.7e-08** [3.9e-08]	9.7e-08*** [1.9e-08]	9.2e-08*** [2.1e-08]
Log Distance $\times$ Water Filtration	.03189 [.06791]	-.029 [.05263]	-.2531*** [.07965]	-.2698*** [.07819]	-.3851*** [.08539]	-.3684*** [.09188]	-.5407*** [.07934]	-.5625*** [.08255]
Log Transportation Cost - Donaldson Hornbeck	.1359 [.2772]	-.369* [.2172]	.3607 [.2585]	-.2351 [.2688]	.883*** [.1846]	.6476*** [.2138]	.6365*** [.1311]	.606*** [.1562]
High-skilled Service	-.02805 [.01933]	-.03666* [.02126]	-.1293*** [.01785]	-.09441*** [.01833]	-.02782* [.01539]	-.02995* [.01686]	-.00551 [.01129]	-.00375 [.01261]
Low-skilled Service	.06796** [.03018]	.07423** [.02959]	-.2033*** [.03351]	-.1382*** [.03388]	.02205 [.03006]	.06747*** [.02235]	.04882** [.02188]	.0756*** [.01919]
Farmers and laborers	-.02925* [.01737]	-.00625 [.01692]	.02293 [.02016]	.00352 [.02053]	.01491 [.01702]	.00061 [.01518]	.00457 [.01122]	-.00169 [.01182]
N/A or unemployed	-.08607*** [.02184]	-.04468** [.01762]	.1382*** [.0391]	.09862*** [.03708]	-.03353* [.01734]	-.04127*** [.01446]	-.0403*** [.01198]	-.03772*** [.01282]
Share White (D-O)		1.196 [.9554]		3.149*** [.8396]		-15.6 [9.719]		-26.93* [15.4]
Share Male (D-O)		-4.366 [3.456]		7.117** [3.226]		82.6*** [15.61]		50.59*** [10.88]
Share Manuf. (D-O)		-4.242** [1.937]		-3.347*** [1.002]		-36.69*** [6.78]		-31.06** [14.09]
Share Foreign Born (D-O)		1.754** [.811]		-1.685** [.7585]		65.49*** [8.343]		31.95*** [11.14]
Share Literacy (D-O)		-6.927** [2.882]		-15.2*** [2.436]		264*** [31.94]		135.1*** [38.53]
Constant	-3.951*** [.2815]	-3.604*** [.2333]	-3.442*** [.2507]	-3.078*** [.2566]	-4.7*** [.1889]	-7.981*** [.5578]	-4.127*** [.1365]	-4.937*** [.2683]
R2	.3034	.3242	.2263	.2355	.4029	.4028	.4741	.4548
F-Stat	3.725	7.008	29	26.44	12.5	180.3	35.49	25.6
Dep. Var.: Mean	-3.8	-3.845	-3.069	-3.215	-3.8	-3.845	-3.069	-3.215
Dep. Var.: Std. Dev.	3.192	3.187	3.641	3.664	3.192	3.187	3.641	3.664
N Obs.	2.5e+05	2.0e+05	5.6e+05	4.6e+05	2.5e+05	2.0e+05	5.6e+05	4.6e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

B. Rural origins and city destinations

Table E.3: Placebo test, dependent variable: migrations 10 years ago

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	.01428 [.5569]	.4191 [.3471]			.4384 [.3398]	.6666* [.3775]		
Typhoid rate (deaths by thousand)	.2355 [1.054]	-.2518 [.9195]						
News Coverage $\times$ Water Filtration			-.00147 [.2266]	-.06773 [.2862]			.1333 [.1945]	-.01761 [.2296]
Water Filtration			-.2882 [.1963]	-.01781 [.2602]				
Log Distance	-.6624*** [.1629]	-.6477*** [.1972]	-.1238 [.1398]	.08426 [.1421]	-.7671*** [.1151]	-.6688*** [.1473]	.07975 [.07619]	.06477 [.09294]
Mean wage of group g in d	2.5e-08 [6.6e-08]	2.0e-08 [7.3e-08]	-2.0e-07** [8.4e-08]	-1.1e-07 [9.1e-08]	8.6e-08* [4.7e-08]	9.1e-08* [5.5e-08]	9.6e-08*** [3.0e-08]	9.7e-08*** [3.6e-08]
Log Distance $\times$ Water Filtration	.1474 [.0913]	.117 [.1424]	-.239 [.156]	-.1577 [.178]	-.3461* [.1761]	-.3251 [.2124]	-.6513*** [.1664]	-.5687*** [.2025]
Log Transportation Cost - Donaldson Hornbeck	1.164*** [.4193]	1.259** [.5573]	.9144** [.3646]	.526 [.409]	1.842*** [.2677]	1.747*** [.3638]	1.187*** [.2212]	1.252*** [.2839]
High-skilled Service	-.03072 [.03341]	-.02953 [.04308]	-.2655*** [.03108]	-.201*** [.03556]	-.0122 [.02453]	-.00411 [.02848]	-.04178** [.02095]	-.03472 [.02433]
Low-skilled Service	-.01552 [.04587]	-.00913 [.06129]	-.2684*** [.03938]	-.1825*** [.04737]	-.05181 [.05347]	.00919 [.0373]	-.05053 [.03874]	-.00299 [.03585]
Farmers and laborers	-.0798*** [.02639]	-.06826** [.03207]	-.03422 [.02553]	-.05992** [.02911]	-.05918** [.02331]	-.0774*** [.02559]	-.105*** [.0181]	-.1121*** [.02055]
N/A or unemployed	-.1073*** [.0297]	-.09251*** [.03443]	.01431 [.04589]	-.02955 [.04715]	-.1064*** [.02269]	-.1202*** [.02244]	-.1666*** [.0189]	-.1692*** [.01999]
Share White (D-O)		2.169 [2.653]		7.23*** [1.35]		30.71 [24.77]		-74.33*** [7.654]
Share Male (D-O)		4.966 [7.675]		14.17*** [4.151]		43.92 [36.62]		50.03 [30.58]
Share Manuf. (D-O)		-4.258 [3.888]		-3.128** [1.588]		-56.45*** [2.966]		-34.46*** [3.706]
Share Foreign Born (D-O)		-1.458 [1.224]		-6.955*** [1.019]		141.4*** [19.56]		25.1*** [7.699]
Share Literacy (D-O)		-17.43*** [5.356]		-24.22*** [3.548]		541*** [66.09]		166.6*** [24.11]
Constant	-4*** [.4438]	-3.943*** [.6893]	-3.349*** [.3693]	-2.628*** [.4356]	-4.741*** [.2885]	-15.87*** [1.686]	-3.996*** [.2451]	-5.24*** [.4319]
R2	.2852	.2889	.2181	.2557	.3778	.4145	.4396	.4736
F-Stat	4.944	6.017	17.21	27.9	18.7	1347	29.84	1097
Dep. Var.: Mean	-3.163	-3.3	-2.393	-2.432	-3.163	-3.3	-2.393	-2.432
Dep. Var.: Std. Dev.	2.911	3.019	3.348	3.432	2.911	3.019	3.348	3.432
N Obs.	83774	58504	1.7e+05	1.2e+05	83774	58504	1.7e+05	1.2e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table E.4: Placebo test, dependent variable: migrations 20 years ago

	Origin-Year-Demographic Group FE				Orig-Year-Group & Dest-Year FE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-.2638 [.8717]	.2949 [.649]			.5583 [.5013]	.6702 [.5113]		
Typhoid rate (deaths by thousand)	.09452 [.8247]	.1928 [.738]						
News Coverage $\times$ Water Filtration			.05484 [.2889]	-.171 [.343]			.1285 [.2563]	-.1335 [.2826]
Water Filtration			-.5917** [.2339]	-.04986 [.2088]				
Log Distance	.06129 [.2004]	.4135** [.2084]	.2254 [.1417]	.7535*** [.1428]	-.1786 [.1432]	-.00995 [.1819]	.2899*** [.07169]	.4078*** [.09069]
Mean wage of group g in d	6.2e-08 [8.7e-08]	-2.7e-08 [8.2e-08]	-2.4e-07*** [8.4e-08]	-1.6e-07* [8.1e-08]	2.9e-08 [5.4e-08]	-1.8e-08 [6.9e-08]	3.0e-08 [3.1e-08]	1.5e-08 [3.8e-08]
Log Distance $\times$ Water Filtration	-.0497 [.1167]	-.2196* [.115]	-.3266** [.1532]	-.4669*** [.1557]	-.7254*** [.1702]	-.7432*** [.1937]	-.7089*** [.16]	-.7675*** [.1819]
Log Transportation Cost - Donaldson Hornbeck	.3613 [.5075]	-.4887 [.4713]	.4856 [.3547]	-.5503 [.4076]	1.387*** [.297]	1.075*** [.3732]	.9187*** [.2027]	.8352*** [.2716]
High-skilled Service	-.03423 [.03604]	-.03115 [.04358]	-.1335*** [.03211]	-.03647 [.03661]	-.03548 [.02935]	-.04143 [.03661]	.00368 [.0227]	.03474 [.02944]
Low-skilled Service	-.07805 [.05435]	-.08337 [.0613]	-.197*** [.04396]	-.08555* [.04604]	-.1153** [.05517]	-.08951* [.05196]	-.03868 [.04067]	.0083 [.04107]
Farmers and laborers	-.0293 [.03278]	-.00629 [.03677]	.03499 [.02846]	.00728 [.03024]	-.001 [.0325]	-.02308 [.03214]	-.02202 [.02187]	-.02846 [.02466]
N/A or unemployed	-.08026** [.0353]	-.04916 [.03927]	.1127** [.04609]	.0537 [.044]	-.05401* [.03031]	-.0672** [.03245]	-.05639** [.02223]	-.05303** [.02585]
Share White (D-O)		1.854 [2.054]		4.371*** [1.217]		37.98*** [7.019]		-44.6*** [7.813]
Share Male (D-O)		-4.724 [5.38]		6.749 [4.323]		44.42*** [15.86]		40.92** [18.29]
Share Manuf. (D-O)		-2.595 [2.839]		-1.761 [1.315]		-36.96*** [1.341]		-20.49*** [2.678]
Share Foreign Born (D-O)		-.2121 [1.09]		-3.68*** [.921]		106.5*** [4.506]		19.7*** [6.449]
Share Literacy (D-O)		-13.81*** [4.363]		-17.45*** [3.032]		409.4*** [15.41]		135.7*** [18.11]
Constant	-3.621*** [.4895]	-2.708*** [.5336]	-3.109*** [.3639]	-2.157*** [.4245]	-4.689*** [.3027]	-12.73*** [.5442]	-3.85*** [.2166]	-4.954*** [.4044]
R2	.2448	.2406	.2368	.2431	.3661	.3464	.4524	.4094
F-Stat	1.486	3.957	24	29.54	9.675	.	38.54	261
Dep. Var.: Mean	-3.082	-3.014	-2.349	-2.499	-3.082	-3.014	-2.349	-2.499
Dep. Var.: Std. Dev.	3.285	3.3	3.507	3.533	3.285	3.3	3.507	3.533
N Obs.	82657	57859	1.7e+05	1.2e+05	82657	57859	1.7e+05	1.2e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table E.5: OD and year FE

	(1)	(2)	(3)	(4)
News Coverage $\times$ Typhoid rate (deaths by thousand)	-.03656 [.0994]	-.1595** [.0397]		
Typhoid rate (deaths by thousand)				
News Coverage $\times$ Water Filtration			.00258 [.01993]	-.01954 [.03159]
Water Filtration				
Mean wage of group g in d	-1.4e-07** [3.4e-08]	-1.5e-07** [4.8e-08]	-9.0e-08** [2.2e-08]	-9.7e-08** [3.0e-08]
Log Distance $\times$ Water Filtration	-.1355*** [.01469]	-.1296*** [.01767]	-.1601*** [.0252]	-.1638*** [.02053]
Log Transportation Cost - Donaldson Hornbeck	-1.834*** [.2657]	-2.166*** [.242]	-1.884*** [.2686]	-2.153*** [.2394]
High-skilled Service	.01036** [.00227]	.01212** [.00306]	.00702* [.00273]	.00889** [.002]
Low-skilled Service	.1726*** [.0167]	.1604*** [.01702]	.1825*** [.01739]	.1695*** [.01573]
Farmers and laborers	.03436*** [.00431]	.03645*** [.00327]	.03292*** [.00269]	.03355*** [.00192]
N/A or unemployed	.01483** [.0043]	.01605*** [.00311]	.01491** [.00356]	.01535*** [.00272]
Share White, O $\times$ Share White, D		1.414** [.4124]		1.104** [.3019]
Share White, D		-1.416* [.512]		-1.107** [.348]
Share Male, O $\times$ Share Male, D		-13.86* [5.614]		-13.33* [5.338]
Share Male, D		7.596 [5.224]		7.973 [4.403]
Share Manuf., O $\times$ Share Manuf.,D		.3628 [1.417]		-2.622 [1.883]
Share Manuf.,D		-.6952 [.3988]		-.5153*** [.0988]
Share Foreign Born, O $\times$ Share Foreign Born, D		-.2519 [1.65]		-.1799 [1.055]
Share Foreign Born, D		1.477 [.9529]		1.719** [.4446]
Share Literate, O $\times$ Share Literate, D		1.638 [1.205]		2.065** [.6645]
Share Literate, D		6.873 [3.718]		7.29** [1.867]
Constant	-1.772** [.4013]	-9.807 [4.677]	-2.315*** [.3823]	-11.61*** [2.48]
R2	.8233	.8284	.8521	.8588
F-Stat	1037	.	541.4	.
Dep. Var.: Mean	-4.562	-4.548	-4.985	-4.976
Dep. Var.: Std. Dev.	1.529	1.542	1.677	1.694
N Obs.	5.5e+05	4.3e+05	1.1e+06	8.5e+05

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

F. Data Appendix

A. IPUMS Definition of Geographical Units

- **URBAN** indicates whether a household's location was urban or rural. It is one of the most used geographic variables and one of the most problematic. Definitions of "urban" vary from year to year - see the background and comparability sections - but the term generally denotes all cities and incorporated places of 2,500+ inhabitants. (All areas not classified as urban are designated rural.)
- **CITY** identifies the city of residence for households located in identifiable cities. This variable is essentially comparable across years, but not all cities are identified in all years, and there are some variations in the exact correspondence between CITY codes and city residents. A year-by-year discussion follows. All rank and size measurements refer to contemporary (not current) population figures, unless otherwise noted:
 - 1850 and 1880: The city of residence is given if the household was in one of the nation's 98 largest cities.
 - 1860 and 1870: The city of residence is given for households in any city with 10,000+ inhabitants.
 - 1900 - 1930: The city of residence is given for households in any city with 25,000+ inhabitants.
- **CITYPOP** reports the population, in hundreds, for all identifiable cities.
- **URBPOP** gives the population, in hundreds, of places considered "urban" according to the census Bureau's 1930-1940 definition. All incorporated municipalities with a population of 2,500 or more were defined as urban, but this definition of urban also included some places that were not incorporated municipalities (See URBAN for a discussion of how the census Bureau's definition changed over time).